ChIP normalization

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1 Introduction

2 Simple model

Consider cells from two **conditons**: one control (vehicle), and one treated with a drug that decreases the abundance of a target epitope. Assume equal amounts of cells, antibody, and spike-in (exogenous) chromatin are used for IP across conditions. Consider the molar ratio in the input of the endogenous target epitope (A), the exogenous target epitope from the spike-in (B), off-target endogenous epitopes (C), and off-target exogenous epitopes from the spike-in (D):

- Control conditon: A:B:C:D
- Treatment condition: A':B:C:D

Assume that antibody-epitope binding reaches equilibrium with dissociation constants of K_{d1} and K_{d2} for target and off-target antibody-epitope interactions, respectively, so $K_{d1} \ll K_{d2}$.

Let Y be antibody concentration. Let Y_f refer to free antibody, while Y_A, Y_B, Y_C, and Y_D refer to antibody-antigen complexes.

Key question: **What are the molar ratios of the epitopes after immunoprecipitation under different regimes of antibody concentration? What is the ratio of spike-in to endogenous epitopes in the immunoprecipitation?**

2.1 Basic equations

Because the endogenous and spike-in target epitopes have the same dissociation constant, the mass-action relation applies to each target population separately:

$$K_{d1} = \frac{Y_f A_f}{Y_A} = \frac{Y_f B_f}{Y_B} = \frac{(Y_f)(A_f + B_f)}{Y_A + Y_B}$$
$$K_{d2} = \frac{Y_f C_f}{Y_C} = \frac{Y_f D_f}{Y_D} = \frac{(Y_f)(C_f + D_f)}{Y_C + Y_D}$$

Below, we will take advantage of the ratios

$$\frac{Y_A}{A_f} = \frac{Y_f}{K_{d1}}, \qquad \frac{Y_B}{B_f} = \frac{Y_f}{K_{d1}}.$$

Conservation of mass:

$$\begin{aligned} A &= A_f + Y_A \\ B &= B_f + Y_B \\ C &= C_f + Y_C \\ D &= D_f + Y_D \\ Y &= Y_f + Y_A + Y_B + Y_C + Y_D \end{aligned}$$

These equations can be used to derive expressions for the bound fractions of each class in terms of the concentration of free antibody Y_f and the dissociation constants K_{d1} and K_{d2}.

$$\begin{aligned} \frac{Y_A}{A} &= \frac{Y_A}{A_f + Y_A} \\ &= \frac{Y_A/A_f}{1 + Y_A/A_f} \\ &= \frac{Y_f/K_{d1}}{1 + Y_f/K_{d1}} \\ &= \frac{Y_f}{Y_f + K_{d1}} \end{aligned}$$

By the same logic,

$$\begin{aligned} \frac{Y_B}{B} &= \frac{Y_f}{Y_f + K_{d1}} \\ \frac{Y_C}{C} &= \frac{Y_f}{Y_f + K_{d2}} \\ \frac{Y_D}{D} &= \frac{Y_f}{Y_f + K_{d2}} \end{aligned}$$

In summary, the proportion bound of each epitope class is

$$\theta_1 = \frac{Y_A}{A} = \frac{Y_B}{B} = \frac{Y_f}{Y_f + K_{d1}}$$
$$\theta_2 = \frac{Y_C}{C} = \frac{Y_D}{D} = \frac{Y_f}{Y_f + K_{d2}}$$

K_{d1} is therefore the concentration of free antibody at equilibrium when half of the target epitopes are bound. Similarly, K_{d2} is the concentration of free antibody at equilibrium when half of the off-target epitopes are bound.

The immunoprecipitation contains amounts $A\theta_1$, $B\theta_1$, $C\theta_2$, and $D\theta_2$. We can simplify this ratio to $A : B : \rho C : \rho D$, where

$$\rho = \frac{\theta_2}{\theta_1} = \frac{Y_f + K_{d1}}{Y_f + K_{d2}}.$$

ρ is the ratio of off-target capture efficiency to target capture efficiency and ranges from $r = \frac{K_{d1}}{K_{d2}} \ll 1$ when antibody is limiting ($Y_f \ll K_{d1}$) to 1 when antibody is in excess ($Y_f \gg K_{d2}$). Below, we use ρ to denote this ratio for the control condition and ρ' to denote this ratio for the treatment condition.

Additional observations:

- The ratios $A : B$ and $C : D$ are always conserved from input to immunoprecipitation.
- The ratio of endogenous epitope to spike-in epitope is
$$\frac{A + \rho C}{B + \rho D}.$$
- The fraction of the immunoprecipitation from the spike-in is
$$\frac{B + \rho D}{A + B + \rho C + \rho D}.$$
As ρ increases from r to 1, this spike-in fraction can either increase or decrease, depending on the sign of $AD - BC$. To see this, differentiate with respect to ρ using the quotient rule:
$$\begin{aligned}\frac{d}{d\rho}\left(\frac{B + \rho D}{A + B + \rho(C + D)}\right) &= \frac{D\big(A + B + \rho(C + D)\big) - (B + \rho D)(C + D)}{\big(A + B + \rho(C + D)\big)^2} \\ &= \frac{AD + BD + \rho CD + \rho D^2 - BC - BD - \rho CD - \rho D^2}{\big(A + B + \rho(C + D)\big)^2} \\ &= \frac{AD - BC}{\big(A + B + \rho(C + D)\big)^2}.\end{aligned}$$
The denominator is positive and the numerator does not depend on ρ, so the spike-in fraction changes monotonically in ρ:

- If $AD > BC$, i.e., $D/B > C/A$, it increases.
- If $AD < BC$, i.e., $D/B < C/A$, it decreases.
- If $AD = BC$, it is constant.

In words, the spike-in fraction increases with ρ if and only if the spike-in has more off-target epitope per target epitope than the cells do.

Regimes. Because $\rho = (Y_f + K_{d1})/(Y_f + K_{d2})$ depends only on the free-antibody concentration Y_f, the regimes below are classified by where the antibody goes. Antibody conservation, $Y = Y_f + S\theta_1 + (C + D)\theta_2$ with $S = A + B$ (or $A' + B$), splits the antibody among three "sinks": solution, target epitopes (capacity S), and off-target epitopes (capacity $C + D$). Each regime corresponds to a different sink dominating or saturating:

- Regime 1 (limiting antibody): target epitopes absorb nearly all antibody.
- Regime 2 (dilute epitopes): epitopes are too dilute to absorb much antibody, so $Y_f \approx Y$.
- Regime 3 (antibody in excess of target): target epitopes are saturated, and the excess antibody is free or bound to unsaturated off-target epitopes.
- Regime 4 (saturating antibody): target and off-target epitopes are both saturated.

The boundaries between these regimes (antibody–target equivalence, off-target half-saturation, and $K_{d1} \sim S$) are treated in the crossovers section. A general result proved there is that $\rho' \geq \rho$ whenever $A' < A$, in every regime: less target leaves more free antibody, which raises off-target capture relative to target capture. The table summarizes the results; the last column is the ChIP-Rx bias factor $(B + \rho D)/(B + \rho' D)$ derived in Regime 1, by which total spike-in normalization multiplies the true fold change a_i'/a_i.

Regime	**Conditions** (for $S = A + B$ and $A' + B$)	$\boldsymbol{\rho}$ **(control)**	$\boldsymbol{\rho'}$ **(treatment)**	$\boldsymbol{\rho'}$ **vs.** $\boldsymbol{\rho}$ **when** $\boldsymbol{A' < A}$	**ChIP-Rx bias**
1. Limiting antibody	$Y < S$; $K_{d1}Y \ll (S-Y)^2$; $rY \ll S - Y$; $r(C+D)Y \ll (S-Y)^2$	$\approx r\dfrac{A+B}{A+B-Y}$	$\approx r\dfrac{A'+B}{A'+B-Y}$	$\rho' > \rho$; both $\to r$ if $Y \ll S$	< 1; ≈ 1 if $\rho' D \ll B$
2. Dilute epitopes	$S \ll K_{d1}$; $C + D \ll K_{d2}$; any Y	$\approx \dfrac{Y+K_{d1}}{Y+K_{d2}}$	$\approx \rho$	$\rho' \approx \rho$ (relative difference $\leq S_c/K_{d1} + (C+D)/K_{d2}$)	≈ 1, even if $\rho D \gtrsim B$
3. Antibody in excess of target	$Y > S$; $rS(K_{d2}+C+D) \ll (Y-S)^2$; $Y - S \ll K_{d2} + C + D$	$\approx r + \dfrac{Y-A-B}{K_{d2}+C+D}$	$\approx r + \dfrac{Y-A'-B}{K_{d2}+C+D}$	$\rho' > \rho$; $\rho' - \rho \approx \dfrac{A-A'}{K_{d2}+C+D}$, not suppressed by r	< 1, possibly far from 1 (overstates the decrease)
4. Saturating antibody	$Y - S - (C+D) \gg K_{d2}$	≈ 1	≈ 1	$\rho' > \rho$, but negligibly	≈ 1, but no enrichment over off-target (IP $\approx$ input)
Crossover: equivalence (1↔3)	$\lvert S-Y\rvert \lesssim$ a few $\sqrt{K_{d1}S}$; $K_{d1} \ll S$	$\approx \dfrac{Y_f+K_{d1}}{K_{d2}}$, $Y_f \approx \sqrt{K_{d1}S}$ at $Y = S$	same with $S = A' + B$	$\rho' \gg \rho$ possible	< 1; largest bias of any regime
Crossover: off-target half-saturation (3↔4)	Target saturated; $Y_f \sim K_{d2}$	$\approx \dfrac{Y_f}{Y_f+K_{d2}}$; $= \frac{1}{2}$ at $Y = S + K_{d2} + \frac{C+D}{2}$	same with $S = A' + B$	$\rho' > \rho$, slightly	≤ 1; ≈ 1 unless $A - A' \sim K_{d2}$

2.2 Regime 1: limiting antibody

Let $S_c = A + B$ and $S_t = A' + B$ be the total target-epitope concentrations in the control and treatment conditions, and let $S \in \{S_c, S_t\}$ denote the total target-epitope concentration of either condition. We assume the following for both conditions (i.e., for each $S \in \{S_c, S_t\}$).

The assumptions are easiest to interpret through a heuristic estimate of the free-antibody concentration. If nearly all antibody is bound to target, then about Y target epitopes are bound and about $S - Y$ are free. Rearranging the mass-action relation $K_{d1} = Y_f(A_f + B_f)/(Y_A + Y_B)$ then gives

$$Y_f = K_{d1}\frac{Y_A + Y_B}{A_f + B_f} \approx K_{d1}\frac{Y}{S-Y}.$$

Below, we show that this estimate is in fact an exact upper bound on Y_f.

1. Antibody is less abundant than target epitope:

$$Y < S.$$

   Thus, even if every antibody molecule binds a target epitope, the remaining target concentration $S - Y$ is positive.

2. Free antibody is scarce relative to the remaining target:

$$K_{d1}Y \ll (S-Y)^2.$$

Dividing both sides by $S-Y$ shows that this is equivalent to $K_{d1}Y/(S-Y) \ll S-Y$: the estimated free antibody is much less than the free target. This matters because antibody that is free does not occupy a target epitope, so it leaves an extra epitope free; the free target concentration stays close to $S-Y$ only if Y_f is small compared with $S-Y$. The square arises because $S-Y$ appears twice: once in the estimate of Y_f (more free target pulls more antibody out of solution) and once as the yardstick against which Y_f is compared.

Taking square roots, the condition is equivalent to $\sqrt{K_{d1}Y} \ll S-Y$. Here $\sqrt{K_{d1}Y}$ is roughly the free-antibody concentration at exact equivalence ($S = Y$): there, the bound target is approximately Y and the free target is $S-(Y-Y_f) = Y_f$, so the mass-action relation gives $K_{d1} \approx Y_f \cdot Y_f/Y$, i.e., $Y_f \approx \sqrt{K_{d1}Y}$. The condition thus says that the excess target $S-Y$ must greatly exceed the free antibody that would remain if the antibody exactly titrated the target.

Formally, below we show that $Y_f \leq K_{d1}Y/(S-Y)$. Dividing both sides by $S-Y$ (which is positive by assumption 1) and then applying this condition,

$$\frac{Y_f}{S-Y} \leq \frac{K_{d1}Y}{(S-Y)(S-Y)} = \frac{K_{d1}Y}{(S-Y)^2} \ll 1.$$

Multiplying both sides by $S-Y$ gives $Y_f \ll S-Y$.

3. The free-antibody concentration is small relative to K_{d2}:

$$rY \ll S-Y.$$

Dividing both sides by $S-Y$ and substituting $r = K_{d1}/K_{d2}$,

$$\frac{rY}{S-Y} = \frac{1}{K_{d2}} \cdot K_{d1}\frac{Y}{S-Y} \ll 1,$$

so this condition says that the estimated free antibody is much less than K_{d2}: off-target epitopes are far from half-occupied. In this unsaturated regime, $\theta_2 = Y_f/(Y_f+K_{d2}) \approx Y_f/K_{d2}$, so $rY/(S-Y)$ is approximately the fraction of off-target epitopes that are bound.

The quantity rY is not especially meaningful on its own; the rearrangement $Y \ll (S-Y)/r$ is more interpretable. Rearranging the mass-action relation, the target-bound antibody is $Y_A+Y_B = Y_f(A_f+B_f)/K_{d1}$. For Y_f to reach K_{d2} while about $S-Y$ target remains free, the target-bound antibody would have to be about $K_{d2}(S-Y)/K_{d1} = (S-Y)/r$. The condition says that the total antibody is far below this amount. Below, we show that this condition implies $Y_f \ll K_{d2}$.

4. Off-target binding consumes little antibody:

$$r(C+D)Y \ll (S-Y)^2.$$

By the mass-action relations, a free target epitope is bound by antibody in the ratio $(Y_A + Y_B)/(A_f+B_f) = Y_f/K_{d1}$, while an off-target epitope is bound in the ratio $(Y_C+Y_D)/(C_f+$

$D_f) = Y_f/K_{d2}$. In the unsaturated regime (assumption 3), nearly all off-target epitopes are free, so $C_f + D_f \approx C + D$ and

$$Y_C + Y_D \approx \frac{Y_f}{K_{d2}}(C + D) = \frac{Y_f}{K_{d1}} \cdot r(C + D).$$

That is, the off-target pool binds as much antibody as $r(C + D)$ free target epitopes would. Thus $r(C + D)$ is the off-target pool measured in "free-target-epitope equivalents."

Dividing both sides of the condition by $S - Y$ and substituting $r = K_{d1}/K_{d2}$,

$$(C + D) \cdot \frac{1}{K_{d2}} \cdot K_{d1} \frac{Y}{S - Y} \ll S - Y.$$

The left-hand side is $C + D$ times the estimated off-target occupancy Y_f/K_{d2}, i.e., the estimated off-target-bound antibody. As in assumption 2, the condition says that antibody that is not on a target epitope (here, antibody bound off-target) is much less than the free target, and the square arises for the same reason. Below, we show that this condition makes the off-target-bound antibody amount negligible relative to the remaining target $S - Y$.

In summary, assumptions 2 and 4 keep the free target concentration close to $S - Y$ by limiting the two places antibody can go other than a target epitope: solution and off-target epitopes. Assumption 3 keeps off-target binding unsaturated.

Simpler sufficient conditions. Suppose $Y \leq S/2$. Multiplying both sides by 2 gives $2Y \leq S$, and subtracting Y from both sides gives $Y \leq S - Y$. Since $S - Y > 0$ by assumption 1, dividing both sides by $S - Y$ gives

$$\frac{Y}{S - Y} \leq 1.$$

Now write the left-hand sides of assumptions 2 and 4, divided by $(S - Y)^2$, as products:

$$\frac{K_{d1}Y}{(S-Y)^2} = \frac{K_{d1}}{S-Y} \cdot \frac{Y}{S-Y} \leq \frac{K_{d1}}{S-Y}, \qquad \frac{r(C+D)Y}{(S-Y)^2} = \frac{r(C+D)}{S-Y} \cdot \frac{Y}{S-Y} \leq \frac{r(C+D)}{S-Y},$$

where each inequality uses $Y/(S - Y) \leq 1$. Therefore, if $K_{d1} \ll S - Y$, then $K_{d1}Y/(S - Y)^2 \ll 1$, which is assumption 2; and if $r(C+D) \ll S-Y$, then $r(C+D)Y/(S-Y)^2 \ll 1$, which is assumption 4. When Y approaches S, however, the factor $Y/(S - Y)$ grows without bound, and the simpler conditions are no longer sufficient.

We now derive ρ for either condition. (For the treatment condition, θ_1, θ_2, Y_f, and ρ should be read as their primed counterparts.)

Antibody conservation. The target-bound antibody is $Y_A + Y_B = A\theta_1 + B\theta_1 = S\theta_1$, and the off-target-bound antibody is $Y_C + Y_D = C\theta_2 + D\theta_2 = (C + D)\theta_2$. Substituting these into $Y = Y_f + Y_A + Y_B + Y_C + Y_D$ gives

$$Y = Y_f + S\theta_1 + (C + D)\theta_2.$$

Step 1: free target. The free target concentration is $S - S\theta_1 = S(1 - \theta_1)$. Solving antibody conservation for $S\theta_1$ gives $S\theta_1 = Y - Y_f - (C + D)\theta_2$, so

$$S(1 - \theta_1) = S - \big(Y - Y_f - (C + D)\theta_2\big) = (S - Y) + Y_f + (C + D)\theta_2. \tag{1}$$

Intuitively, the free target is the target left over if all antibody were bound to target $(S - Y)$, plus the target "freed up" by antibody that is free or bound off-target.

Step 2: an exact expression for ρ. From $\theta_1 = Y_f/(Y_f + K_{d1})$,

$$1 - \theta_1 = \frac{Y_f + K_{d1}}{Y_f + K_{d1}} - \frac{Y_f}{Y_f + K_{d1}} = \frac{K_{d1}}{Y_f + K_{d1}},$$

so, multiplying the numerator and denominator by S,

$$Y_f + K_{d1} = \frac{K_{d1}}{1 - \theta_1} = \frac{K_{d1} S}{S(1 - \theta_1)}. \tag{2}$$

Substituting (2) into the numerator of ρ, and then (1) into the result,

$$\begin{aligned}
\rho &= \frac{Y_f + K_{d1}}{Y_f + K_{d2}} \\
&= \frac{K_{d1} S}{S(1 - \theta_1)\,(Y_f + K_{d2})} \\
&= \frac{K_{d1} S}{\big[(S - Y) + Y_f + (C + D)\theta_2\big](K_{d2} + Y_f)}.
\end{aligned}$$

Factoring $S - Y$ out of the first bracket and K_{d2} out of the second, and using $r = K_{d1}/K_{d2}$,

$$\rho = r \frac{S}{S - Y} \cdot \frac{1}{(1 + \varepsilon_1)(1 + \varepsilon_2)}, \qquad \varepsilon_1 = \frac{Y_f + (C + D)\theta_2}{S - Y}, \quad \varepsilon_2 = \frac{Y_f}{K_{d2}}. \tag{3}$$

Equation (3) is exact. It remains to show that $\varepsilon_1 \ll 1$ and $\varepsilon_2 \ll 1$.

Step 3: an upper bound on Y_f. Because $Y_f \geq 0$ and $(C + D)\theta_2 \geq 0$, equation (1) gives $S(1 - \theta_1) \geq S - Y$, which is positive by assumption 1. A smaller denominator gives a larger fraction, so substituting into (2),

$$Y_f + K_{d1} \leq \frac{K_{d1} S}{S - Y}.$$

Subtracting K_{d1} from both sides,

$$Y_f \leq \frac{K_{d1} S}{S - Y} - \frac{K_{d1}(S - Y)}{S - Y} = \frac{K_{d1} Y}{S - Y}. \tag{4}$$

Step 4: $\varepsilon_2 \ll 1$. Dividing (4) by K_{d2},

$$\varepsilon_2 = \frac{Y_f}{K_{d2}} \leq \frac{K_{d1}}{K_{d2}} \cdot \frac{Y}{S - Y} = \frac{rY}{S - Y} \ll 1, \tag{5}$$

where the final inequality is assumption 3. Therefore, $Y_f \ll K_{d2}$.

Step 5: $\varepsilon_1 \ll 1$. Split ε_1 into its two terms. For the free-antibody term, dividing (4) by $S - Y$,

$$\frac{Y_f}{S - Y} \leq \frac{K_{d1} Y}{(S - Y)^2} \ll 1,$$

by assumption 2. For the off-target term, dropping Y_f from the denominator of θ_2 makes it larger, so by (5),

$$\theta_2 = \frac{Y_f}{Y_f + K_{d2}} \leq \frac{Y_f}{K_{d2}} \leq \frac{rY}{S - Y}.$$

Multiplying by $(C + D)/(S - Y)$,

$$\frac{(C + D)\theta_2}{S - Y} \leq \frac{r(C + D)Y}{(S - Y)^2} \ll 1,$$

by assumption 4. Since both terms are $\ll 1$, so is their sum ε_1.

Step 6: result. Since $\varepsilon_1 \ll 1$ and $\varepsilon_2 \ll 1$, $(1 + \varepsilon_1)(1 + \varepsilon_2) \approx 1$, and (3) becomes

$$\rho \approx r\frac{S}{S - Y}.$$

(Because $\varepsilon_1, \varepsilon_2 \geq 0$, (3) also shows that $r\,S/(S - Y)$ is an exact upper bound on ρ.) Substituting $S = A + B$ for the control and $S = A' + B$ for the treatment,

$$\rho \approx r\frac{A + B}{A + B - Y}, \qquad \rho' \approx r\frac{A' + B}{A' + B - Y}.$$

If, in addition, $Y \ll \min(A + B, A' + B)$, these reduce to $\rho \approx \rho' \approx r$.

Note that ρ' **increases relative to** ρ **when** $A' < A$. This can be seen from the expression

$$\rho \approx r\frac{S}{S - Y} = r\frac{S - Y + Y}{S - Y} = r\left(1 + \frac{Y}{S - Y}\right).$$

For $S > Y$ (assumption 1), decreasing S decreases the positive denominator $S - Y$, so $\frac{Y}{S-Y}$ and hence ρ increase.

2.2.1 ChIP normalization

In the control condition, the ratio of endogenous to spike-in epitopes in the immunoprecipitation is $\frac{A+\rho C}{B+\rho D}$

In the treatment condition, the ratio of endogenous to spike-in epitopes in the immunoprecipitation is $\frac{A'+\rho' C}{B+\rho' D}$.

Goal. Let a_i and a_i' be the concentrations of target epitope at genomic locus i in the control and treatment conditions, so that $\sum_i a_i = A$ and $\sum_i a_i' = A'$. We want normalized signals whose ratio between conditions at each locus is the true fold change a_i'/a_i.

Reads. Assume that sequencing samples reads in proportion to the composition of the immunoprecipitation, and that target loci do not overlap off-target loci. Let N be the total number of reads in the control library. The immunoprecipitation contains $a_i\theta_1$ at locus i out of a total of

$(A+B)\theta_1+(C+D)\theta_2$. Dividing the numerator and denominator by θ_1 and substituting $\rho = \theta_2/\theta_1$, the expected number of reads at locus i is

$$n_i = N\frac{a_i\theta_1}{(A+B)\theta_1+(C+D)\theta_2} = N\frac{a_i}{A+B+\rho(C+D)}.$$

By the same logic, the expected number of reads mapping to the spike-in genome (from both target and off-target spike-in epitopes) is

$$N_s = N\frac{B+\rho D}{A+B+\rho(C+D)}.$$

The treatment library has the analogous quantities N', n'_i, and N'_s, with A, a_i, and ρ replaced by A', a'_i, and ρ'.

Scaling by total reads (RPM). Multiplying each library by $\alpha = 1/N$ (or $10^6/N$, in reads per million) gives

$$\frac{n_i}{N} = \frac{a_i}{A+B+\rho(C+D)}, \qquad \frac{n'_i}{N'} = \frac{a'_i}{A'+B+\rho'(C+D)}.$$

The ratio between conditions is

$$\frac{n'_i/N'}{n_i/N} = \frac{a'_i}{a_i}\cdot\frac{A+B+\rho(C+D)}{A'+B+\rho'(C+D)}.$$

If the treatment decreases the target globally ($A' < A$) and the spike-in and off-target terms are small, the second factor is approximately $A/A' > 1$, which cancels the global decrease. Thus, RPM normalization hides global changes; this is the motivation for spike-in normalization.

Scaling by spike-in reads (ChIP-Rx). Multiplying each library by $\alpha = 1/N_s$ for the control and $\alpha' = 1/N'_s$ for the treatment (in practice, $10^6/N_s$ and $10^6/N'_s$), the factors N and $A+B+\rho(C+D)$ cancel:

$$\frac{n_i}{N_s} = \frac{N\,a_i/(A+B+\rho(C+D))}{N\,(B+\rho D)/(A+B+\rho(C+D))} = \frac{a_i}{B+\rho D}, \qquad \frac{n'_i}{N'_s} = \frac{a'_i}{B+\rho' D}.$$

The ratio between conditions is

$$\frac{n'_i/N'_s}{n_i/N_s} = \frac{a'_i}{a_i}\cdot\frac{B+\rho D}{B+\rho' D}.$$

The second factor is a bias. It equals 1 if $\rho D \ll B$ and $\rho' D \ll B$, i.e., if off-target binding to the spike-in is negligible compared with target binding to the spike-in. In the limiting-antibody regime, substituting the approximations for ρ and ρ', this requires

$$r\frac{A+B}{A+B-Y}D \ll B, \qquad r\frac{A'+B}{A'+B-Y}D \ll B.$$

Otherwise, because $\rho' > \rho$ when $A' < A$, we have $B+\rho' D > B+\rho D$, so the bias factor is less than 1: spike-in normalization *overstates* the decrease in the treatment condition.

Scaling by spike-in target reads. The bias arises because N_s includes off-target spike-in reads, whose capture efficiency differs between conditions. If instead we count only the reads $N_{s,B}$ at the spike-in's target loci (e.g., reads in spike-in peaks), then by the same logic as above,

$$N_{s,B} = N\frac{B}{A+B+\rho(C+D)}, \qquad N'_{s,B} = N'\frac{B}{A'+B+\rho'(C+D)}.$$

Using scaling factors $\alpha = 1/N_{s,B}$ and $\alpha' = 1/N'_{s,B}$,

$$\frac{n_i}{N_{s,B}} = \frac{a_i}{B}, \qquad \frac{n'_i}{N'_{s,B}} = \frac{a'_i}{B}, \qquad \frac{n'_i/N'_{s,B}}{n_i/N_{s,B}} = \frac{a'_i}{a_i},$$

which recovers the true fold change regardless of ρ and ρ'. (In practice, reads in spike-in peaks also include some off-target background, so this cancellation is only approximate.)

2.2.2 Example: Pol II S2ph ChIP in K562 cells with an mESC spike-in

We now ask what antibody and spike-in amounts would place a realistic experiment in Regime 1. All parameter values below are order-of-magnitude estimates; the point is the method, which can be rerun with better numbers.

Experimental setup.

- 5 million K562 cells per condition; control is vehicle, and the treatment (e.g., a CDK9 inhibitor) halves Ser2-phosphorylated Pol II, so $A' = A/2$.
- Spike-in: mouse embryonic stem cell (mESC) chromatin from 5×10^5 cells (10% of the K562 cell number).
- Antibody: Cell Signaling Technology #13499, Phospho-Rpb1 CTD (Ser2) (E1Z3G) rabbit monoclonal antibody (IgG, ≈ 150 kDa). The manufacturer recommends 10 μL (1:50) per IP of 10 μg chromatin ($\approx 4 \times 10^6$ cells). The stock concentration is lot-specific and not listed on the product page; we assume 50 μg/mL, so 10 μL is ≈ 0.5 μg.
- IP volume $V = 500$ μL (typical for SimpleChIP-style protocols).

Epitope abundances. A HeLa cell contains roughly 3×10^5 Pol II molecules, of which roughly 0.65–1.1×10^5 are hyperphosphorylated (Pol IIO) or actively transcribing (Kimura et al., 1999; Mol Cell Biol 19:5383). K562 cells are somewhat larger than HeLa cells with substantially larger nuclei, so we take 10^5 Ser2-phosphorylated Pol II per K562 cell. We are not aware of a direct count in mESCs; given their smaller size, we take 5×10^4 per mESC. We treat each Ser2-phosphorylated Pol II as a single target epitope, even though the CTD contains 52 heptad repeats that can each carry Ser2 phosphorylation. Multiple epitopes per molecule mainly increase the effective (avidity-enhanced) affinity, which we absorb into K_{d1}. For off-target epitopes (non-S2ph Pol II, other phosphoproteins, and nonspecific chromatin sites), we assume 10^7 sites per cell in both species.

To convert a number of molecules per cell into a concentration in the IP, multiply by the number of cells, divide by Avogadro's number $N_A = 6.022 \times 10^{23}$ mol^{-1}, and divide by V:

$$A = \frac{(5 \times 10^6)(10^5)}{(6.022 \times 10^{23})(5 \times 10^{-4}\ \text{L})} = \frac{5 \times 10^{11}}{3.011 \times 10^{20}}\ \text{M} \approx 1.66\ \text{nM},$$

$$B = \frac{(5 \times 10^5)(5 \times 10^4)}{(6.022 \times 10^{23})(5 \times 10^{-4}\ \text{L})} \approx 0.083\ \text{nM},$$

$$C = \frac{(5 \times 10^6)(10^7)}{(6.022 \times 10^{23})(5 \times 10^{-4}\ \text{L})} \approx 166\ \text{nM}, \qquad D \approx 16.6\ \text{nM}.$$

With $A' = A/2 \approx 0.83$ nM, the total target concentrations are

$$S_c = A + B \approx 1.74 \text{ nM}, \qquad S_t = A' + B \approx 0.91 \text{ nM}, \qquad C + D \approx 183 \text{ nM}.$$

Affinities. We assume an effective target affinity $K_{d1} = 0.1$ nM (typical of rabbit monoclonal antibodies, and helped by avidity) and an off-target affinity $K_{d2} = 1\ \mu\text{M} = 1000$ nM, so $r = K_{d1}/K_{d2} = 10^{-4}$.

The recommended antibody amount is not in Regime 1. An IgG of 150 kDa has $1/(1.5 \times 10^5 \text{ g/mol}) \approx 6.7$ pmol per μg. Thus 0.5 μg in 500 μL is

$$Y = \frac{0.5 \times 10^{-6} \text{ g}}{(1.5 \times 10^5 \text{ g/mol})(5 \times 10^{-4} \text{ L})} \approx 6.7 \text{ nM},$$

which exceeds both S_c and S_t, violating assumption 1. The recommended protocol therefore operates with antibody in excess of target (Regime 3 below).

Choosing Y for Regime 1. The treatment condition, with the smaller S, is the binding constraint, so we check each assumption with $S = S_t \approx 0.91$ nM. Try $Y = 0.2$ nM, so $S_t - Y \approx 0.71$ nM.

1. $Y = 0.2 \text{ nM} < S_t \approx 0.91$ nM. ✓
2. $\dfrac{K_{d1} Y}{(S_t - Y)^2} = \dfrac{(0.1)(0.2)}{(0.71)^2} \approx 0.04 \ll 1.$ ✓
3. $\dfrac{rY}{S_t - Y} = \dfrac{(10^{-4})(0.2)}{0.71} \approx 3 \times 10^{-5} \ll 1.$ ✓
4. $\dfrac{r(C + D)Y}{(S_t - Y)^2} = \dfrac{(10^{-4})(183)(0.2)}{(0.71)^2} \approx 0.007 \ll 1.$ ✓

In the control, $S_c - Y \approx 1.54$ nM is larger, so every ratio is even smaller. Assumption 2 is the tightest: increasing Y to 0.45 nM ($\approx S_t/2$) would raise its ratio to $(0.1)(0.45)/(0.46)^2 \approx 0.2$, which is no longer $\ll 1$.

Converting $Y = 0.2$ nM into an amount of antibody,

$$(0.2 \times 10^{-9} \text{ M})(5 \times 10^{-4} \text{ L}) = 10^{-13} \text{ mol}, \qquad (10^{-13} \text{ mol})(1.5 \times 10^5 \text{ g/mol}) = 1.5 \times 10^{-8} \text{ g} = 15 \text{ ng},$$

i.e., about 0.3 μL of the assumed 50 μg/mL stock, roughly 30-fold less than recommended. (If both arms of the IgG bind independently, the binding-site concentration is $2Y$, and the antibody amount should be halved again.)

Predicted capture efficiencies and bias. Substituting into the Regime 1 result,

$$\rho \approx r \frac{S_c}{S_c - Y} = 10^{-4} \cdot \frac{1.74}{1.54} \approx 1.13 \times 10^{-4},$$
$$\rho' \approx r \frac{S_t}{S_t - Y} = 10^{-4} \cdot \frac{0.91}{0.71} \approx 1.28 \times 10^{-4}.$$

The ChIP-Rx bias factor is then

$$\frac{B+\rho D}{B+\rho' D} = \frac{0.083 + (1.13\times 10^{-4})(16.6)}{0.083 + (1.28\times 10^{-4})(16.6)} = \frac{0.0849}{0.0851} \approx 0.997,$$

so total-spike-in normalization is accurate to $\approx 0.3\%$. For comparison, solving the full conservation equation numerically at the recommended $Y \approx 6.7$ nM gives $\rho \approx 4.3\times 10^{-3}$, $\rho' \approx 5.0\times 10^{-3}$, and a bias factor of ≈ 0.93: ChIP-Rx would overstate the treatment-induced decrease by $\approx 7\%$, and off-target epitopes would make up nearly half ($\rho D/(B+\rho D) \approx 0.46$) of the spike-in reads.

Yield. In Regime 1 nearly all antibody is bound to target, so about $Y \cdot V = 10^{-13}$ mol $\approx 6\times 10^{10}$ Pol II molecules are captured: $\theta_1 \approx Y/S$ is about 11% of target epitopes in the control and 22% in the treatment. At $\sim$200 bp per captured fragment and 650 g/mol per base pair, this is roughly $(10^{-13})(200)(650) \approx 13$ ng of DNA, before losses, which is ample for library preparation.

Practical notes. If the antibody amount cannot be reduced this far, increasing S helps: concentrating the IP into a smaller volume or using more cells raises $S-Y$ for a fixed amount of antibody, which relaxes assumption 2 quadratically. Because the spike-in contributes only $\approx 5\%$ of the target ($B/A \approx 0.05$), changing the spike-in amount barely affects whether Regime 1 holds. Its main role is to ensure enough spike-in target reads, and to keep $\rho D \ll B$. Finally, the stock concentration, K_{d1}, and the epitope counts should be measured or checked (e.g., against the lot's certificate of analysis) before relying on these numbers.

2.3 Regime 2: dilute epitopes (no antibody depletion)

As in Regime 1, let $S_c = A + B$ and $S_t = A' + B$ be the total target-epitope concentrations in the control and treatment conditions, and let $S \in \{S_c, S_t\}$ denote either one. In Regime 1, antibody was the scarce reagent: nearly every antibody molecule found a target, so the amount of target captured was fixed at about Y and the *fraction* of target captured depended on S. Here we consider the opposite situation: the epitopes are so dilute that binding them removes only a negligible amount of antibody from solution. The free antibody concentration is then set by the experimenter ($Y_f \approx Y$) rather than by the chromatin, and, as we will see, every capture efficiency becomes independent of S.

We assume the following for both conditions (i.e., for each $S \in \{S_c, S_t\}$).

1. Target epitopes are dilute relative to K_{d1}:

$$S \ll K_{d1}.$$

   To interpret the ratio S/K_{d1}, rearrange the mass-action relation $K_{d1} = Y_f(A_f+B_f)/(Y_A+Y_B)$ to give the ratio of target-bound to free antibody:

$$\frac{Y_A+Y_B}{Y_f} = \frac{A_f+B_f}{K_{d1}}.$$

   When the target is unsaturated, nearly all target is free ($A_f + B_f \approx S$), so S/K_{d1} is approximately the ratio of target-bound antibody to free antibody. The assumption therefore says

that for every antibody molecule sitting on a target epitope, many more are free in solution. (If the target is partly saturated, fewer target epitopes are free, $A_f + B_f < S$, so S/K_{d1} is an upper bound on this ratio.) Note the contrast with Regime 1, which required the opposite inequality, $K_{d1} \ll S - Y$.

2. Off-target epitopes are dilute relative to K_{d2}:

$$C + D \ll K_{d2}.$$

By the same logic applied to the off-target mass-action relation,

$$\frac{Y_C + Y_D}{Y_f} = \frac{C_f + D_f}{K_{d2}} \leq \frac{C + D}{K_{d2}},$$

so $(C + D)/K_{d2}$ bounds the ratio of off-target-bound to free antibody. Because $K_{d2} \gg K_{d1}$, this assumption is usually much easier to satisfy than assumption 1: the off-target pool may be far larger than the target pool and still bind little antibody.

In summary, the two assumptions say that the two places antibody can go other than solution—target and off-target epitopes—each hold a small amount of antibody compared with the amount that stays free. Note that neither assumption involves Y: in this regime, the antibody may be limiting, intermediate, or in excess relative to the dissociation constants; it is only required to be in excess of what the epitopes can soak up. Because $S_t < S_c$ when $A' < A$, the control condition is the more demanding one, so it suffices to check the assumptions with $S = S_c$.

We now derive Y_f, θ_1, and ρ for either condition. (For the treatment condition, θ_1, θ_2, Y_f, and ρ should be read as their primed counterparts.)

Antibody conservation. As derived in Regime 1, substituting $Y_A + Y_B = S\theta_1$ and $Y_C + Y_D = (C + D)\theta_2$ into $Y = Y_f + Y_A + Y_B + Y_C + Y_D$ gives

$$Y = Y_f + S\theta_1 + (C + D)\theta_2. \tag{6}$$

Step 1: bound the antibody on epitopes. Dropping Y_f from the denominator of $\theta_1 = Y_f/(Y_f + K_{d1})$ makes the denominator smaller and hence the fraction larger:

$$\theta_1 = \frac{Y_f}{Y_f + K_{d1}} \leq \frac{Y_f}{K_{d1}}.$$

Multiplying both sides by S,

$$S\theta_1 \leq S\frac{Y_f}{K_{d1}} = \frac{S}{K_{d1}}\, Y_f.$$

By the same logic with $\theta_2 = Y_f/(Y_f + K_{d2}) \leq Y_f/K_{d2}$,

$$(C + D)\theta_2 \leq \frac{C + D}{K_{d2}}\, Y_f.$$

Step 2: $Y_f \approx Y$. Define the relative amount of epitope-bound antibody

$$\delta = \frac{S\theta_1 + (C + D)\theta_2}{Y_f},$$

so that (6) reads $Y = Y_f + \delta Y_f = Y_f(1+\delta)$. Adding the two inequalities from Step 1 and dividing by Y_f,

$$0 \le \delta \le \frac{S}{K_{d1}} + \frac{C+D}{K_{d2}} \equiv \bar{\delta} \ll 1, \tag{7}$$

where $\delta \ge 0$ because every term is nonnegative, and the final $\ll$ follows from assumptions 1 and 2 (a sum of two small numbers is small). Solving $Y = Y_f(1+\delta)$ for Y_f and using $1 \le 1+\delta \le 1+\bar{\delta}$,

$$\frac{Y}{1+\bar{\delta}} \le Y_f = \frac{Y}{1+\delta} \le Y. \tag{8}$$

Since $\bar{\delta} \ll 1$, $Y_f \approx Y$: the free antibody is essentially all of the antibody. (Equivalently, the fraction of antibody that is bound, $(Y - Y_f)/Y = \delta/(1+\delta)$, is at most $\bar{\delta}$.)

Step 3: capture efficiencies. Substituting $Y_f \approx Y$ into the expressions for the bound fractions,

$$\theta_1 \approx \frac{Y}{Y + K_{d1}}, \qquad \theta_2 \approx \frac{Y}{Y + K_{d2}}.$$

Neither depends on S. In particular, the fraction of target epitope captured is the same in both conditions, so the IP yield *per epitope* does not change when the treatment removes target. (The total target captured, $S\theta_1$, is proportional to S and so does decrease.) This is the key difference from Regime 1, where the total captured was fixed at about Y and the per-epitope efficiency $\theta_1 \approx Y/S$ rose when S fell.

Step 4: ρ. Substituting $Y_f \approx Y$ into $\rho = (Y_f + K_{d1})/(Y_f + K_{d2})$ gives

$$\rho \approx \rho' \approx \bar{\rho} \equiv \frac{Y + K_{d1}}{Y + K_{d2}}. \tag{9}$$

Since $\bar{\rho}$ depends only on Y, K_{d1}, and K_{d2}—not on S—it is the same for the control and treatment conditions, regardless of A'.

To make "$\approx$" precise, we bound ρ above and below. First, ρ increases with Y_f: by the quotient rule,

$$\frac{d\rho}{dY_f} = \frac{(Y_f + K_{d2}) - (Y_f + K_{d1})}{(Y_f + K_{d2})^2} = \frac{K_{d2} - K_{d1}}{(Y_f + K_{d2})^2} > 0,$$

since $K_{d2} > K_{d1}$. Because $Y_f \le Y$ by (8), it follows that $\rho \le \bar{\rho}$. For the lower bound, substitute the lower bound $Y_f \ge Y/(1+\bar{\delta})$ from (8) and multiply the numerator and denominator by $1+\bar{\delta}$:

$$\rho \ge \frac{\dfrac{Y}{1+\bar{\delta}} + K_{d1}}{\dfrac{Y}{1+\bar{\delta}} + K_{d2}} = \frac{Y + (1+\bar{\delta})K_{d1}}{Y + (1+\bar{\delta})K_{d2}}.$$

The numerator is at least $Y + K_{d1}$, and the denominator is at most $(1+\bar{\delta})Y + (1+\bar{\delta})K_{d2} = (1+\bar{\delta})(Y + K_{d2})$, so

$$\rho \ge \frac{Y + K_{d1}}{(1+\bar{\delta})(Y + K_{d2})} = \frac{\bar{\rho}}{1+\bar{\delta}}.$$

Together,

$$\frac{\bar{\rho}}{1+\bar{\delta}} \le \rho \le \bar{\rho}, \tag{10}$$

so ρ agrees with $\bar{\rho}$ to within a relative error of at most $\bar{\delta} \ll 1$. The same holds for ρ' (with $\bar{\delta}$ computed using S_t, which is smaller than with S_c).

Direction of the (small) difference. Although $\rho' \approx \rho$, the exact values still move in the same direction as in Regime 1. Write the right-hand side of (6) as a function of Y_f and S:

$$g(Y_f, S) = Y_f + S\frac{Y_f}{Y_f + K_{d1}} + (C + D)\frac{Y_f}{Y_f + K_{d2}}.$$

Each term increases with Y_f (for instance, $\frac{d}{dY_f}\frac{Y_f}{Y_f+K} = \frac{K}{(Y_f+K)^2} > 0$), and g increases with S. If $A' < A$, then $S_t < S_c$, so

$$g(Y_f, S_t) < g(Y_f, S_c) = Y = g(Y'_f, S_t).$$

Since $g(\cdot, S_t)$ is increasing, this requires $Y_f < Y'_f$: with less target to bind, slightly more antibody stays free. Because ρ increases with Y_f, $\rho < \rho'$. Combining with (10) for both conditions,

$$\rho < \rho' \leq \bar{\rho} \leq (1 + \bar{\delta})\rho, \tag{11}$$

where $\bar{\delta}$ is evaluated for the control ($S = S_c$). Thus ρ' exceeds ρ, but by a relative amount of at most $\bar{\delta} \ll 1$.

Sub-limits of $\bar{\rho}$. Factoring K_{d1} out of the numerator and K_{d2} out of the denominator of (9),

$$\bar{\rho} = \frac{K_{d1}(1 + Y/K_{d1})}{K_{d2}(1 + Y/K_{d2})} = r\,\frac{1 + Y/K_{d1}}{1 + Y/K_{d2}}.$$

- $Y \ll K_{d1}$ (antibody below the target K_d): both $Y/K_{d1} \ll 1$ and $Y/K_{d2} \ll 1$, so $\bar{\rho} \approx r$. Both epitope classes are unsaturated, $\theta_1 \approx Y/K_{d1}$ and $\theta_2 \approx Y/K_{d2}$, and their ratio is set purely by the affinities.
- $K_{d1} \ll Y \ll K_{d2}$: in the numerator, $Y + K_{d1} \approx Y$; in the denominator, $Y + K_{d2} \approx K_{d2}$. So $\bar{\rho} \approx Y/K_{d2}$. The target is saturated ($\theta_1 \approx 1$), while off-target binding is still in its linear regime ($\theta_2 \approx Y/K_{d2}$), so off-target capture grows in proportion to Y while target capture cannot grow further.
- $Y \gg K_{d2}$: both $Y + K_{d1} \approx Y$ and $Y + K_{d2} \approx Y$, so $\bar{\rho} \approx 1$. Both classes are saturated and the antibody no longer discriminates between them.

As Y increases, $\bar{\rho}$ rises monotonically from r to 1 (by the same derivative computed above, with Y in place of Y_f).

2.3.1 ChIP normalization

The read-count expressions derived in Regime 1 (expected reads n_i at target locus i and N_s on the spike-in genome) did not use any regime-specific assumptions, so they apply here unchanged.

Scaling by spike-in reads (ChIP-Rx). From Regime 1,

$$\frac{n'_i/N'_s}{n_i/N_s} = \frac{a'_i}{a_i} \cdot \frac{B + \rho D}{B + \rho' D}.$$

Substituting $\rho \approx \rho' \approx \bar{\rho}$ from (9), the numerator and denominator of the bias factor are both approximately $B + \bar{\rho}D$, so

$$\frac{B + \rho D}{B + \rho' D} \approx \frac{B + \bar{\rho} D}{B + \bar{\rho} D} = 1, \qquad \frac{n_i'/N_s'}{n_i/N_s} \approx \frac{a_i'}{a_i}.$$

More precisely, by (11), $\rho D < \rho' D \leq (1 + \bar{\delta})\rho D$. Adding B and using $B \leq (1 + \bar{\delta})B$,

$$B + \rho D < B + \rho' D \leq (1 + \bar{\delta})(B + \rho D),$$

and taking reciprocals and multiplying by $B + \rho D$,

$$\frac{1}{1 + \bar{\delta}} \leq \frac{B + \rho D}{B + \rho' D} < 1.$$

So spike-in normalization recovers the true fold change to within a relative error of at most $\bar{\delta}$, *even if off-target spike-in reads are not negligible* (ρD comparable to or larger than B). This contrasts with Regime 1, where the bias factor equals 1 only if $\rho D \ll B$. The reason is that in this regime the off-target spike-in reads are captured with the same efficiency in both conditions, so they scale the spike-in read count by the same factor in both libraries and cancel in the ratio. (The direction of the residual bias, < 1, is the same as in Regime 1, but its size is bounded by $\bar{\delta}$.) Scaling by spike-in target reads $N_{s,B}$ also works, as it did in Regime 1.

Scaling by total reads (RPM). From Regime 1, substituting $\rho' \approx \rho$,

$$\frac{n_i'/N'}{n_i/N} = \frac{a_i'}{a_i} \cdot \frac{A + B + \rho(C + D)}{A' + B + \rho'(C + D)} \approx \frac{a_i'}{a_i} \cdot \frac{A + B + \rho(C + D)}{A' + B + \rho(C + D)}.$$

If $A' < A$, the numerator of the second factor exceeds its denominator, so the factor is greater than 1 and partially cancels the global decrease (fully, $\approx A/A'$, if the spike-in and off-target terms are small). RPM normalization still hides global changes in this regime.

The cost of this regime. Dilute epitopes make normalization robust, but they come at a price.

- *Low yield when $Y \ll K_{d1}$.* The fraction of target captured is $\theta_1 \approx Y/(Y + K_{d1}) \approx Y/K_{d1} \ll 1$, and the total target captured is $S\theta_1 \approx SY/K_{d1}$, which is small because both $S \ll K_{d1}$ and $Y \ll K_{d1}$. Little DNA is recovered, so libraries have low complexity and are sensitive to contamination.
- *Off-target contamination when Y is raised.* Increasing Y to improve yield raises $\bar{\rho}$ from r toward Y/K_{d2} and eventually 1. Among endogenous reads, the target-to-off-target ratio is $A : \bar{\rho}C$, so the fraction of endogenous reads that are on target,

  $$\frac{A}{A + \bar{\rho}C},$$

  falls as Y grows. Once $Y \gg K_{d1}$, target capture is saturated ($\theta_1 \approx 1$), so further increases in Y only add off-target background. Because $\bar{\rho}$ is the same in both conditions, this background does not bias spike-in normalization at target loci (under the assumption that target and off-target loci do not overlap), but it lowers the signal-to-noise ratio, especially in the treatment condition, where the on-target fraction $A'/(A' + \bar{\rho}C)$ is smaller because $A' < A$.

2.4 Regime 3: antibody in excess of target

As in Regime 1, let $S_c = A + B$ and $S_t = A' + B$ be the total target-epitope concentrations in the control and treatment conditions, and let $S \in \{S_c, S_t\}$ denote either one. We assume the following for both conditions (i.e., for each $S \in \{S_c, S_t\}$).

The assumptions are again easiest to interpret through a heuristic estimate of the free-antibody concentration. Now there is more antibody than target. If nearly every target epitope is bound, then about S antibody molecules sit on target, and the excess $Y - S$ must go somewhere else: either into solution (Y_f) or onto off-target epitopes ($Y_C + Y_D$). If off-target epitopes are far from saturated, nearly all of them are free ($C_f + D_f \approx C + D$), so rearranging the mass-action relation $K_{d2} = Y_f(C_f + D_f)/(Y_C + Y_D)$ gives

$$Y_C + Y_D = \frac{Y_f(C_f + D_f)}{K_{d2}} \approx \frac{Y_f(C + D)}{K_{d2}}.$$

Substituting this into "excess = free + off-target-bound,"

$$Y - S \approx Y_f + \frac{Y_f(C + D)}{K_{d2}} = Y_f \cdot \frac{K_{d2} + C + D}{K_{d2}},$$

and multiplying both sides by $K_{d2}/(K_{d2} + C + D)$,

$$Y_f \approx K_{d2} \frac{Y - S}{K_{d2} + C + D}.$$

In words, the excess antibody is split between solution and off-target epitopes in the ratio $K_{d2} : (C + D)$. Below, we show that this estimate is in fact an exact lower bound on Y_f.

1. Antibody is more abundant than target epitope:

   $$Y > S.$$

   Thus, even if every target epitope is bound, there is a positive excess $Y - S$ of antibody left over.

2. Free target is scarce relative to the excess antibody:

   $$rS\,(K_{d2} + C + D) \ll (Y - S)^2.$$

   To interpret this, estimate the free target concentration $A_f + B_f$. Rearranging the mass-action relation $K_{d1} = Y_f(A_f + B_f)/(Y_A + Y_B)$, and using that about S target epitopes are bound,

   $$A_f + B_f = K_{d1} \frac{Y_A + Y_B}{Y_f} \approx \frac{K_{d1} S}{Y_f}.$$

   Substituting the heuristic estimate of Y_f and then $r = K_{d1}/K_{d2}$,

   $$A_f + B_f \approx \frac{K_{d1} S\,(K_{d2} + C + D)}{K_{d2}(Y - S)} = \frac{rS\,(K_{d2} + C + D)}{Y - S}.$$

Dividing both sides of the condition by $Y - S$ shows that it is equivalent to $rS(K_{d2} + C + D)/(Y - S) \ll Y - S$: the estimated free target is much less than the excess antibody. This matters because every target epitope that is free leaves one more antibody molecule unaccounted for, so the antibody available to go into solution or off-target stays close to $Y - S$ only if the free target is small compared with $Y - S$. Usually this also means that the target is nearly saturated ($\theta_1 \approx 1$). The square arises because $Y - S$ appears twice: once in the estimate of the free target (more excess antibody drives more target into the bound state) and once as the yardstick against which the free target is compared.

Mirror symmetry with Regime 1. Substituting $r = K_{d1}/K_{d2}$ and splitting the fraction, the left-hand side is

$$rS\left(K_{d2} + C + D\right) = \frac{K_{d1}}{K_{d2}} S\left(K_{d2} + C + D\right) = K_{d1} S\left(1 + \frac{C + D}{K_{d2}}\right).$$

If there were no off-target epitopes ($C + D = 0$), the condition would read $K_{d1}S \ll (Y - S)^2$. Compare Regime 1's assumption 2, $K_{d1}Y \ll (S - Y)^2$: the two are identical after swapping Y and S. This is because the mass-action relation $K_{d1} = Y_f(A_f + B_f)/(Y_A + Y_B)$ treats antibody and target symmetrically. In Regime 1, antibody is the limiting species and target is in excess; in Regime 3, the roles are reversed. In both cases the condition says that the leftover *limiting* species (free antibody in Regime 1, free target here) is much less than the excess of the *abundant* species. The extra factor $1 + (C + D)/K_{d2}$ breaks the symmetry: off-target epitopes are a sink for antibody but not for target, so only a fraction $K_{d2}/(K_{d2} + C + D)$ of the excess antibody remains free to drive target binding. A weaker push on the target requires a larger excess to compensate.

Formally, below we show that $S(1 - \theta_1) \leq rS(K_{d2} + C + D)/(Y - S)$, so this condition makes the free target negligible relative to $Y - S$.

3. Off-target epitopes are far from saturated:

$$Y - S \ll K_{d2} + C + D.$$

By the heuristic estimate, the off-target occupancy is

$$\theta_2 = \frac{Y_f}{Y_f + K_{d2}} \approx \frac{Y_f}{K_{d2}} \approx \frac{Y - S}{K_{d2} + C + D},$$

so this condition says that the fraction of off-target epitopes that are bound is small. Another way to see the yardstick $K_{d2} + C + D$: to half-occupy the off-target epitopes, the free antibody must reach K_{d2} and about $(C + D)/2$ antibody molecules must be bound off-target, for a total of about $K_{d2} + (C + D)/2$, which is within a factor of 2 of $K_{d2} + C + D$. The condition says that the excess antibody is far below this amount. Below, we show that this condition implies $Y_f \ll K_{d2}$.

In summary, assumption 2 ensures that essentially all of the excess antibody $Y - S$ is available to the "non-target" compartments (solution and off-target epitopes), and assumption 3 keeps the off-target epitopes in their unsaturated, linear-response regime. Compared with Regime 1, there is no separate assumption limiting off-target binding: here, off-target binding is not a perturbation to be neglected but one of the two places where the excess antibody goes.

Simpler sufficient condition. Suppose $Y \geq 2S$. Subtracting S from both sides gives $Y - S \geq S$, and since $Y - S > 0$ by assumption 1, dividing both sides by $Y - S$ gives

$$\frac{S}{Y - S} \leq 1.$$

Writing the left-hand side of assumption 2, divided by $(Y - S)^2$, as a product,

$$\frac{rS\,(K_{d2} + C + D)}{(Y - S)^2} = \frac{r(K_{d2} + C + D)}{Y - S} \cdot \frac{S}{Y - S} \leq \frac{r(K_{d2} + C + D)}{Y - S}.$$

Therefore, if $r(K_{d2} + C + D) \ll Y - S$, then assumption 2 holds. When Y approaches S, however, the factor $S/(Y - S)$ grows without bound, and the simpler condition is no longer sufficient.

We now derive ρ for either condition. (For the treatment condition, θ_1, θ_2, Y_f, and ρ should be read as their primed counterparts.)

Antibody conservation. As in Regime 1,

$$Y = Y_f + S\theta_1 + (C + D)\theta_2.$$

Step 1: ρ in terms of θ_2. Substituting $K_{d1} = rK_{d2}$ into the numerator of ρ, and then adding and subtracting rY_f,

$$\begin{aligned}\rho &= \frac{Y_f + K_{d1}}{Y_f + K_{d2}} = \frac{Y_f + rK_{d2}}{Y_f + K_{d2}} \\ &= \frac{r(Y_f + K_{d2}) + Y_f - rY_f}{Y_f + K_{d2}} \\ &= r + (1 - r)\frac{Y_f}{Y_f + K_{d2}}.\end{aligned}$$

Recognizing $\theta_2 = Y_f/(Y_f + K_{d2})$,

$$\rho = r + (1 - r)\,\theta_2. \tag{12}$$

This is exact and holds in every regime. It says that ρ interpolates linearly between r (no off-target occupancy) and 1 (full off-target occupancy). So it remains to find θ_2.

Step 2: where the excess antibody goes. Write the target-bound antibody as total target minus free target, $S\theta_1 = S - S(1 - \theta_1)$. Substituting into antibody conservation,

$$Y = Y_f + S - S(1 - \theta_1) + (C + D)\theta_2.$$

Subtracting S from both sides and adding $S(1 - \theta_1)$ to both sides,

$$(Y - S) + S(1 - \theta_1) = Y_f + (C + D)\theta_2. \tag{13}$$

Intuitively, the antibody not on target (right-hand side) equals the excess antibody $Y - S$ plus one antibody for each target epitope that is left free. This is the mirror image of Regime 1's equation (1).

Step 3: an exact expression for θ_2. From $\theta_2 = Y_f/(Y_f + K_{d2})$, multiplying both sides by $Y_f + K_{d2}$ gives $Y_f = \theta_2(Y_f + K_{d2})$. Substituting this into the right-hand side of (13) and factoring out θ_2,

$$(Y - S) + S(1 - \theta_1) = \theta_2(Y_f + K_{d2}) + (C + D)\theta_2 = \theta_2\,(K_{d2} + C + D + Y_f).$$

Dividing both sides by $K_{d2} + C + D + Y_f$,

$$\theta_2 = \frac{(Y - S) + S(1 - \theta_1)}{K_{d2} + C + D + Y_f}.$$

Factoring $Y - S$ out of the numerator and $K_{d2} + C + D$ out of the denominator,

$$\theta_2 = \frac{Y - S}{K_{d2} + C + D} \cdot \frac{1 + \delta_1}{1 + \delta_2}, \qquad \delta_1 = \frac{S(1 - \theta_1)}{Y - S}, \quad \delta_2 = \frac{Y_f}{K_{d2} + C + D}. \tag{14}$$

Equations (12) and (14) together are exact. Here δ_1 is the free target relative to the excess antibody, and δ_2 is the free antibody relative to $K_{d2} + C + D$. It remains to show that $\delta_1 \ll 1$ and $\delta_2 \ll 1$.

Step 4: a lower bound on Y_f. Because $S(1 - \theta_1) \geq 0$, equation (13) gives

$$Y_f + (C + D)\theta_2 \geq Y - S.$$

Dropping Y_f from the denominator of θ_2 makes it larger, so $\theta_2 = Y_f/(Y_f + K_{d2}) \leq Y_f/K_{d2}$. Substituting this on the left-hand side (which can only make the left-hand side larger, so the inequality still holds),

$$Y_f + \frac{(C + D)Y_f}{K_{d2}} \geq Y - S \quad \Longrightarrow \quad Y_f \cdot \frac{K_{d2} + C + D}{K_{d2}} \geq Y - S.$$

Multiplying both sides by the positive number $K_{d2}/(K_{d2} + C + D)$,

$$Y_f \geq K_{d2}\frac{Y - S}{K_{d2} + C + D}, \tag{15}$$

which is positive by assumption 1. This is the heuristic estimate from the start of this subsection, now shown to be a lower bound.

Step 5: $\delta_1 \ll 1$. From $\theta_1 = Y_f/(Y_f + K_{d1})$, as in Regime 1, $1 - \theta_1 = K_{d1}/(Y_f + K_{d1})$. Dropping K_{d1} from the denominator makes the fraction larger, and then substituting the lower bound (15) for Y_f in the denominator makes it larger still:

$$1 - \theta_1 = \frac{K_{d1}}{Y_f + K_{d1}} \leq \frac{K_{d1}}{Y_f} \leq \frac{K_{d1}(K_{d2} + C + D)}{K_{d2}(Y - S)} = \frac{r(K_{d2} + C + D)}{Y - S},$$

where the last step substitutes $r = K_{d1}/K_{d2}$. Multiplying both sides by $S/(Y - S)$,

$$\delta_1 = \frac{S(1 - \theta_1)}{Y - S} \leq \frac{rS\,(K_{d2} + C + D)}{(Y - S)^2} \ll 1,$$

by assumption 2.

Step 6: $\theta_2 \ll 1$, $Y_f \ll K_{d2}$, and $\delta_2 \ll 1$. Because $\delta_2 \geq 0$, the factor $1/(1+\delta_2)$ in (14) is at most 1, so

$$\theta_2 \leq \frac{Y-S}{K_{d2}+C+D}\,(1+\delta_1) \ll 1,$$

since $(Y-S)/(K_{d2}+C+D) \ll 1$ by assumption 3 and $1+\delta_1 \approx 1$ by Step 5. To convert this into a statement about Y_f, start from $Y_f = \theta_2(Y_f + K_{d2})$, subtract $\theta_2 Y_f$ from both sides to get $Y_f(1-\theta_2) = \theta_2 K_{d2}$, and divide by $K_{d2}(1-\theta_2)$:

$$\frac{Y_f}{K_{d2}} = \frac{\theta_2}{1-\theta_2}.$$

With $\theta_2 \ll 1$, the right-hand side is approximately $\theta_2 \ll 1$, so $Y_f \ll K_{d2}$. Finally, because $K_{d2} \leq K_{d2}+C+D$,

$$\delta_2 = \frac{Y_f}{K_{d2}+C+D} \leq \frac{Y_f}{K_{d2}} \ll 1.$$

Step 7: result. Since $\delta_1 \ll 1$ and $\delta_2 \ll 1$, $(1+\delta_1)/(1+\delta_2) \approx 1$, and (14) becomes

$$\theta_2 \approx \frac{Y-S}{K_{d2}+C+D}.$$

Substituting this into (12), and using $1-r \approx 1$ (because $K_{d1} \ll K_{d2}$),

$$\rho \approx r + \frac{Y-S}{K_{d2}+C+D}. \tag{16}$$

Correspondingly, since $Y_f/K_{d2} = \theta_2/(1-\theta_2) \approx \theta_2$, the free antibody is $Y_f \approx K_{d2}(Y-S)/(K_{d2}+C+D)$, confirming the heuristic estimate. Substituting $S = A+B$ for the control and $S = A'+B$ for the treatment,

$$\rho \approx r + \frac{Y-A-B}{K_{d2}+C+D}, \qquad \rho' \approx r + \frac{Y-A'-B}{K_{d2}+C+D}.$$

(An exact lower bound also follows. Since $d\theta_2/dY_f = K_{d2}/(Y_f+K_{d2})^2 > 0$, θ_2 increases with Y_f, so substituting the lower bound (15) into $\theta_2 = Y_f/(Y_f+K_{d2})$ and then multiplying the numerator and denominator by $(K_{d2}+C+D)/K_{d2}$,

$$\theta_2 \geq \frac{K_{d2}\frac{Y-S}{K_{d2}+C+D}}{K_{d2}\frac{Y-S}{K_{d2}+C+D}+K_{d2}} = \frac{Y-S}{(Y-S)+(K_{d2}+C+D)}.$$

Substituting into (12) gives $\rho \geq r + (1-r)(Y-S)/(Y-S+K_{d2}+C+D)$, which agrees with (16) to first order when $Y-S \ll K_{d2}+C+D$.)

Two terms contribute to ρ. The first, r, is the baseline ratio of off-target to target capture efficiency set by the affinities. The second is the fraction of off-target epitopes occupied by the excess antibody. Because target is saturated ($\theta_1 \approx 1$), any increase in off-target occupancy translates directly into an increase in ρ. Unlike in Regime 1, the second term is not multiplied by r, so ρ is not necessarily close to r: it can be many times larger.

Two sub-cases. The denominator $K_{d2}+C+D$ reflects the two places where the excess antibody can go, in the ratio $K_{d2} : (C+D)$.

(3a) Few off-target epitopes, $C + D \ll K_{d2}$. Then $K_{d2} + C + D \approx K_{d2}$, so

$$Y_f \approx K_{d2}\frac{Y - S}{K_{d2}} = Y - S, \qquad \rho \approx r + \frac{Y - S}{K_{d2}}.$$

The excess antibody stays free in solution, and the off-target occupancy is simply Y_f/K_{d2}. Assumption 3 becomes $Y - S \ll K_{d2}$, and assumption 2 becomes $K_{d1}S \ll (Y - S)^2$, the exact mirror of Regime 1's assumption 2.

(3b) Many off-target epitopes, $C + D \gg K_{d2}$. Then $K_{d2} + C + D \approx C + D$, so

$$Y_f \approx K_{d2}\frac{Y - S}{C + D} \ll Y - S, \qquad \rho \approx r + \frac{Y - S}{C + D}.$$

The off-target pool soaks up nearly all of the excess antibody, so the excess divided by the number of off-target epitopes is the fraction of off-target epitopes that are occupied. Assumption 3 becomes $Y - S \ll C + D$.

Chromatin contains an enormous number of weak, nonspecific binding sites, so sub-case (3b) is plausible in practice.

(Remark on assumption 3. The approximation to θ_2 in (14) actually requires only $\delta_2 \ll 1$, i.e., $Y_f \ll K_{d2} + C + D$. In sub-case (3b), this can remain true even when a substantial fraction of off-target epitopes is occupied, because the large off-target pool keeps Y_f small. Then the exact relation (12) with $\theta_2 \approx (Y - S)/(C + D)$ still holds, and ρ continues to grow linearly with the excess until $Y - S$ approaches $C + D$. We keep assumption 3 as stated because it defines the regime in which $\rho \ll 1$.)

How ρ' compares to ρ. Subtracting the approximations for ρ and ρ', the r terms cancel and the fractions share a common denominator:

$$\begin{aligned}\rho' - \rho &\approx \left(r + \frac{Y - S_t}{K_{d2} + C + D}\right) - \left(r + \frac{Y - S_c}{K_{d2} + C + D}\right) \\ &= \frac{(Y - S_t) - (Y - S_c)}{K_{d2} + C + D} = \frac{S_c - S_t}{K_{d2} + C + D}.\end{aligned}$$

Substituting $S_c = A + B$ and $S_t = A' + B$, the spike-in terms B cancel:

$$\rho' - \rho \approx \frac{(A + B) - (A' + B)}{K_{d2} + C + D} = \frac{A - A'}{K_{d2} + C + D}. \tag{17}$$

Since the denominator is positive, $\rho' > \rho$ **when** $A' < A$. Intuitively, the treatment removes $A - A'$ target epitopes, which frees the $A - A'$ antibody molecules that would have occupied them. That antibody joins the excess, and a fraction $(C + D)/(K_{d2} + C + D)$ of it lands on off-target epitopes, raising the off-target occupancy.

The sign of $\rho' - \rho$ is in fact exact. Define $g(Y_f) = Y_f + SY_f/(Y_f + K_{d1}) + (C + D)Y_f/(Y_f + K_{d2})$, so that antibody conservation reads $g(Y_f) = Y$. Each term of g increases with Y_f, and for $Y_f > 0$ the middle term increases with S. If S decreases (from S_c to S_t) while Y is held fixed, then g evaluated at the old Y_f falls below Y, so the new solution must have a larger Y_f. By the quotient rule,

$$\frac{d\rho}{dY_f} = \frac{(Y_f + K_{d2}) - (Y_f + K_{d1})}{(Y_f + K_{d2})^2} = \frac{K_{d2} - K_{d1}}{(Y_f + K_{d2})^2} > 0,$$

so a larger Y_f means a larger ρ. This argument applies in every regime, including Regime 1.

Note that the magnitude (17), unlike the Regime 1 result, is not suppressed by a factor of r. In Regime 1, a decrease in target changes ρ only by changing how tightly the (limiting) antibody is held on target, an effect proportional to r. Here, the decrease in target releases antibody directly into the off-target pool.

Relevance to practice. Typical ChIP protocols use micrograms of antibody. An IgG has a molar mass of about 150 kDa, so 1 μg is about 10^{-6} g/$(1.5 \times 10^5$ g/mol) ≈ 6.7 pmol. By comparison, a factor with 10^4 binding sites per cell in 10^6 cells has 10^{10} sites, or about $10^{10}/(6.0 \times 10^{23}$ mol$^{-1}) \approx$ 0.017 pmol. For many targets, then, standard practice places the IP in this antibody-excess regime. (Very abundant epitopes, such as some histone modifications, may instead be comparable to or exceed the antibody.)

2.4.1 ChIP normalization

The read-count model of Regime 1 applies unchanged, because it used only the composition $A : B : \rho C : \rho D$ of the immunoprecipitation. In particular, ChIP-Rx scaling gives

$$\frac{n_i'/N_s'}{n_i/N_s} = \frac{a_i'}{a_i} \cdot \frac{B + \rho D}{B + \rho' D}.$$

Rewriting the bias factor. Write $\rho' = \rho + (\rho' - \rho)$, so the denominator is $B + \rho' D = (B + \rho D) + (\rho' - \rho)D$. Dividing the numerator and denominator of the bias factor by $B + \rho D$,

$$\frac{B + \rho D}{B + \rho' D} = \frac{B + \rho D}{(B + \rho D) + (\rho' - \rho)D} = \frac{1}{1 + \dfrac{(\rho' - \rho)D}{B + \rho D}}.$$

Since $\rho' > \rho$ when $A' < A$, the fraction in the denominator is positive, so the bias factor is less than 1: as in Regime 1, **ChIP-Rx overstates the decrease** in the treatment condition. The bias factor is approximately 1 only if

$$(\rho' - \rho)D \ll B + \rho D.$$

Substituting the regime expressions. Substituting (17) for $\rho' - \rho$ and (16) for ρ, the bias factor is

$$\frac{B + \rho D}{B + \rho' D} \approx \frac{1}{1 + \dfrac{(A - A')D}{(K_{d2} + C + D)\big(B + \rho D\big)}}, \qquad \rho \approx r + \frac{Y - A - B}{K_{d2} + C + D},$$

and ChIP-Rx is unbiased only if

$$\frac{(A - A')D}{K_{d2} + C + D} \ll B + \left(r + \frac{Y - A - B}{K_{d2} + C + D}\right) D.$$

The left-hand side has a direct interpretation. Splitting the fraction,

$$\frac{(A - A')D}{K_{d2} + C + D} = (A - A') \cdot \frac{C + D}{K_{d2} + C + D} \cdot \frac{D}{C + D} :$$

the $A - A'$ antibody molecules freed by the treatment, times the fraction of excess antibody that is captured off-target, times the share of off-target epitopes that belong to the spike-in. That is the extra antibody that ends up on spike-in off-target epitopes in the treatment. The right-hand side, $B + \rho D = (B\theta_1 + D\theta_2)/\theta_1$, is the spike-in material in the control IP divided by $\theta_1 \approx 1$. So the condition says that the extra spike-in capture caused by the freed antibody must be small compared with the spike-in capture already present.

How large can the bias be? Consider sub-case (3b), so $K_{d2} + C + D \approx C + D$, and suppose off-target spike-in reads are a minor part of the spike-in signal, $\rho D \ll B$. The fraction in the bias factor becomes

$$\frac{(A - A')D}{(K_{d2} + C + D)(B + \rho D)} \approx \frac{(A - A')D}{(C + D)B}.$$

Suppose further that the spike-in chromatin has the same ratio of off-target to target epitopes as the cells, $D/B = C/A$. Substituting $D/B = C/A$,

$$\frac{(A - A')D}{(C + D)B} = \frac{(A - A')}{C + D} \cdot \frac{C}{A} = \frac{A - A'}{A} \cdot \frac{C}{C + D}.$$

If the spike-in is a small fraction of the chromatin ($D \ll C$), then $C/(C + D) \approx 1$ and the fraction is approximately $1 - A'/A$, the fractional loss of target. The bias factor is then

$$\frac{1}{1 + (1 - A'/A)} = \frac{1}{2 - A'/A},$$

and the measured global fold change is

$$\frac{A'}{A} \cdot \frac{1}{2 - A'/A}.$$

For example, if the treatment halves the target ($A'/A = 1/2$), the bias factor is $1/(2 - 1/2) = 2/3$, and ChIP-Rx reports a fold change of $(1/2)(2/3) = 1/3$ instead of $1/2$. This bias is of order 1, whereas in Regime 1 it was suppressed by r. The reason is that in this regime the freed antibody is redistributed almost entirely onto off-target epitopes, and the spike-in receives a share of that redistribution comparable to its share of the target signal. (If instead $\rho D \gg B$, the bias factor approaches $\rho D/(\rho' D) = \rho/\rho'$, the ratio of off-target capture efficiencies, which can also differ substantially from 1 when the excess $Y - S$ is not large compared with $A - A'$.)

Scaling by spike-in target reads. The derivation of Regime 1 used only the read-count model, not any regime-specific approximation. Therefore, counting only reads $N_{s,B}$ at the spike-in's target loci still gives

$$\frac{n_i'/N_{s,B}'}{n_i/N_{s,B}} = \frac{a_i'}{a_i},$$

removing the bias regardless of ρ and ρ'. This matters more here than in Regime 1, because the off-target spike-in reads that N_s includes are exactly what changes between conditions. (As before, reads in spike-in peaks include some off-target background, so the cancellation is only approximate; the residual error is larger when ρD is larger.)

2.5 Regime 4: saturating antibody

In this regime, antibody is so abundant that even the weak, off-target interactions are saturated. As in Regime 1, let $S \in \{S_c, S_t\}$ with $S_c = A + B$ and $S_t = A' + B$, and assume the following for both conditions.

Assumption. The antibody left over after binding *every* epitope, target and off-target, is much greater than K_{d2}:

$$Y - S - (C + D) \gg K_{d2}.$$

For brevity, write $E = Y - S - (C + D)$ for this **excess antibody** (E_c and E_t for the two conditions). Because $S_t < S_c$ when $A' < A$,

$$E_t = Y - S_t - (C + D) > Y - S_c - (C + D) = E_c,$$

so it suffices to check the assumption for the control condition, $E_c \gg K_{d2}$.

Intuition. Each epitope can hold at most one antibody, so at most $S + (C + D)$ antibody can be bound, and at least E must remain free. Recall that K_{d2} is the free-antibody concentration at which off-target epitopes are half-bound. If the guaranteed free antibody E is already far above K_{d2}, then off-target epitopes are nearly all bound—and target epitopes, which bind more tightly ($K_{d1} < K_{d2}$), are even more completely bound. Note that this requires both $Y \gg K_{d2}$ *and* $Y > S + C + D$: antibody must outnumber all epitopes, including the typically very abundant off-target ones.

We now derive ρ for either condition. (For the treatment condition, Y_f, θ_1, θ_2, E, and ρ should be read as their primed counterparts.)

Step 1: a lower bound on Y_f. Start from antibody conservation (derived in Regime 1),

$$Y = Y_f + S\theta_1 + (C + D)\theta_2.$$

Solving for Y_f, and then adding and subtracting $S + (C + D)$,

$$\begin{aligned} Y_f &= Y - S\theta_1 - (C + D)\theta_2 \\ &= \big[Y - S - (C + D)\big] + S - S\theta_1 + (C + D) - (C + D)\theta_2 \\ &= E + S(1 - \theta_1) + (C + D)(1 - \theta_2). \end{aligned}$$

Intuitively, the free antibody is the excess antibody E, plus one antibody "returned to solution" for each epitope that is unbound. Because $\theta_1 \le 1$ and $\theta_2 \le 1$, the last two terms are nonnegative, so

$$Y_f \ge E. \tag{18}$$

By the assumption, $E \gg K_{d2}$, so $Y_f \gg K_{d2}$. Since $K_{d1} < K_{d2}$, also $Y_f \gg K_{d1}$.

Step 2: both classes of epitope are nearly fully bound. From $\theta_2 = Y_f/(Y_f + K_{d2})$,

$$1 - \theta_2 = \frac{Y_f + K_{d2}}{Y_f + K_{d2}} - \frac{Y_f}{Y_f + K_{d2}} = \frac{K_{d2}}{Y_f + K_{d2}}.$$

Dropping K_{d2} from the denominator makes the fraction larger, and then substituting (18) (a larger denominator makes it smaller),

$$1 - \theta_2 \leq \frac{K_{d2}}{Y_f} \leq \frac{K_{d2}}{E} \ll 1.$$

By the same logic, $1 - \theta_1 = K_{d1}/(Y_f + K_{d1}) \leq K_{d1}/E$, which is even smaller because $K_{d1} < K_{d2}$. Thus $\theta_1 \approx \theta_2 \approx 1$: nearly every epitope, target or off-target, is bound by antibody.

Step 3: an exact expression for $1 - \rho$. Writing 1 with the same denominator as ρ,

$$1 - \rho = \frac{Y_f + K_{d2}}{Y_f + K_{d2}} - \frac{Y_f + K_{d1}}{Y_f + K_{d2}} = \frac{K_{d2} - K_{d1}}{Y_f + K_{d2}}. \tag{19}$$

Since $K_{d2} > K_{d1}$, the numerator is positive, so $1 - \rho > 0$, i.e., $\rho < 1$.

Step 4: $\rho \approx 1$. In (19), the numerator satisfies $K_{d2} - K_{d1} < K_{d2}$, and the denominator satisfies $Y_f + K_{d2} > Y_f \geq E$ by (18). A larger numerator and a smaller denominator each make a positive fraction larger, so

$$0 < 1 - \rho < \frac{K_{d2}}{Y_f} \leq \frac{K_{d2}}{E} = \frac{K_{d2}}{Y - S - (C + D)} \ll 1. \tag{20}$$

Therefore $\rho \approx 1$, and by the same argument for the treatment condition, $\rho' \approx 1$.

Step 5: comparing ρ' and ρ. First, ρ increases with Y_f. Differentiating with the quotient rule,

$$\frac{d\rho}{dY_f} = \frac{(1)(Y_f + K_{d2}) - (Y_f + K_{d1})(1)}{(Y_f + K_{d2})^2} = \frac{K_{d2} - K_{d1}}{(Y_f + K_{d2})^2} > 0.$$

Second, $Y_f' > Y_f$ when $S_t < S_c$. To see this, substitute $\theta_1 = Y_f/(Y_f + K_{d1})$ and $\theta_2 = Y_f/(Y_f + K_{d2})$ into antibody conservation, and define the right-hand side as a function of a trial free-antibody concentration $x \geq 0$ and the target total S:

$$F(x, S) = x + S\frac{x}{x + K_{d1}} + (C + D)\frac{x}{x + K_{d2}}.$$

The equilibrium Y_f is the value of x that solves $F(x, S) = Y$.

- F is increasing in x: each term is increasing, since $\frac{d}{dx}x = 1 > 0$ and, by the quotient rule, for any $K > 0$,
$$\frac{d}{dx}\frac{x}{x + K} = \frac{(1)(x + K) - x(1)}{(x + K)^2} = \frac{K}{(x + K)^2} > 0.$$
- F is increasing in S for $x > 0$: $\partial F/\partial S = x/(x + K_{d1}) = \theta_1 > 0$.

The two conditions share Y, so $F(Y_f, S_c) = Y = F(Y_f', S_t)$. Because $S_t < S_c$ and $Y_f > 0$ (by (18)), the second property gives

$$F(Y_f, S_t) < F(Y_f, S_c) = Y = F(Y_f', S_t).$$

Since $F(\cdot, S_t)$ is increasing, $F(Y_f, S_t) < F(Y_f', S_t)$ requires $Y_f < Y_f'$. In words: with fewer target epitopes to absorb antibody in the treatment condition, more antibody is left free.

Combining the two facts, $Y_f' > Y_f$ implies $\rho' > \rho$, just as in Regime 1. But both lie below 1 (Step 3), so using (20),

$$0 < \rho' - \rho < 1 - \rho < \frac{K_{d2}}{E_c} \ll 1.$$

The difference is negligible: $\rho \approx \rho' \approx 1$.

Result. The immunoprecipitation contains $A\theta_1 : B\theta_1 : C\theta_2 : D\theta_2 = A : B : \rho C : \rho D$, so

$$\text{IP composition} \approx A : B : C : D,$$

which is the same as the input. Because nearly every epitope is captured (Step 2), the immunoprecipitation is essentially a copy of the epitope-containing chromatin, with no preference for target over off-target epitopes.

2.5.1 ChIP normalization

We reuse the read model and normalization formulas from Regime 1, substituting $\rho \approx \rho' \approx 1$.

Scaling by spike-in reads (ChIP-Rx). From Regime 1, the ratio between conditions is

$$\frac{n_i'/N_s'}{n_i/N_s} = \frac{a_i'}{a_i} \cdot \frac{B + \rho D}{B + \rho' D}.$$

Since $\rho' > \rho$ (Step 5), $B + \rho' D > B + \rho D$, so the bias factor is slightly less than 1. To bound how far below 1 it can be, use $\rho' < 1$ to enlarge the denominator, $B + \rho' D < B + D$, which makes the fraction smaller:

$$\frac{B + \rho D}{B + \rho' D} > \frac{B + \rho D}{B + D} = \frac{(B + D) - (1 - \rho)D}{B + D} = 1 - (1 - \rho)\frac{D}{B + D}.$$

Since $D/(B + D) \leq 1$ and, by (20), $1 - \rho < K_{d2}/E_c$,

$$1 - \frac{K_{d2}}{E_c} < \frac{B + \rho D}{B + \rho' D} < 1.$$

So the bias factor is ≈ 1, and **ChIP-Rx normalization is essentially unbiased** at target loci in this idealized model: both conditions capture both spike-in epitope classes with the same (nearly complete) efficiency, so the spike-in reads track the amount of chromatin just as they would in the input.

Scaling by total reads (RPM). Substituting $\rho \approx \rho' \approx 1$ into the Regime 1 ratio,

$$\frac{n_i'/N'}{n_i/N} = \frac{a_i'}{a_i} \cdot \frac{A + B + \rho(C + D)}{A' + B + \rho'(C + D)} \approx \frac{a_i'}{a_i} \cdot \frac{A + B + C + D}{A' + B + C + D}.$$

If the off-target pool dominates ($C + D \gg A$), the second factor is close to 1 and even RPM recovers the global change; the large, unchanging off-target pool acts as an internal reference. This is not a virtue, as we now see.

The cost: loss of specificity. Let c_j be the off-target epitope concentration at an off-target locus j. The immunoprecipitation contains $a_i\theta_1$ at target locus i and $c_j\theta_2$ at locus j. Dividing both by θ_1 and substituting $\rho = \theta_2/\theta_1$,

$$\frac{a_i\theta_1}{c_j\theta_2} = \frac{a_i}{\rho\, c_j} \approx \frac{a_i}{c_j},$$

which is the same ratio as in the input. Compare Regime 1, where $\rho \approx r$ boosts target over off-target by a factor $1/r = K_{d2}/K_{d1} \gg 1$. Here that enrichment is lost entirely. Summing over loci, the fraction of IP reads that come from endogenous off-target epitopes is

$$\frac{\rho C}{A + B + \rho(C + D)} \approx \frac{C}{A + B + C + D},$$

and the fraction from all off-target epitopes is approximately $(C + D)/(A + B + C + D)$. When off-target epitopes are far more abundant than the target (as is typical, e.g., for an antibody that weakly recognizes unmodified histones), most reads are off-target and peaks at target loci are diluted into the background. Normalization is unbiased, but the signal it normalizes is mostly noise.

In practice. This regime is an idealization. Real ChIP "background" includes not only weak antibody-epitope binding (which saturates, as modeled here) but also chromatin that sticks non-specifically to beads, tubes, or antibody-bead complexes. That background does not follow the binding isotherm $Y_f/(Y_f + K_{d2})$; it roughly scales with the amount of input chromatin and need not be the same in the endogenous and spike-in chromatin. Moreover, reaching $Y > S+C+D+K_{d2}$ requires enormous amounts of antibody when the off-target pool is large. The main lesson of this regime is therefore not that saturating antibody makes spike-in normalization safe, but that adding antibody beyond the point of saturating the target trades specificity for (at best) uniform capture: the immunoprecipitation increasingly resembles the input.

2.6 Crossovers between regimes

Regimes 1–4 each describe a limit in which one or two terms of antibody conservation dominate. Real experiments need not sit deep inside any one limit, and the boundaries between regimes are exactly where the treatment and control conditions are most likely to behave differently. In this section we first prove a result that holds in *every* regime, and then work out the three boundary regions explicitly:

- the **equivalence crossover** ($Y \approx S$), between regimes 1 and 3;
- the **off-target half-saturation crossover** ($Y_f \sim K_{d2}$), between regimes 3 and 4;
- the **weak-binding crossover** ($K_{d1} \sim S$), between regimes 1 and 2.

As before, $S \in \{S_c, S_t\}$ with $S_c = A + B$ and $S_t = A' + B$, and $r = K_{d1}/K_{d2}$. We abbreviate the off-target pool as $M = C + D$, which is the same in both conditions.

2.6.1 A general principle: less target means more free antibody

Step 1: ρ depends only on Y_f, and increases with it. By definition,

$$\rho = \frac{Y_f + K_{d1}}{Y_f + K_{d2}},$$

which involves no concentration other than Y_f. Treating ρ as a function of Y_f and differentiating with the quotient rule,

$$\frac{d\rho}{dY_f} = \frac{1 \cdot (Y_f + K_{d2}) - (Y_f + K_{d1}) \cdot 1}{(Y_f + K_{d2})^2} = \frac{K_{d2} - K_{d1}}{(Y_f + K_{d2})^2}.$$

Since $K_{d1} < K_{d2}$, the numerator is positive, and the denominator is a square, so

$$\frac{d\rho}{dY_f} = \frac{K_{d2} - K_{d1}}{(Y_f + K_{d2})^2} > 0. \tag{21}$$

Thus, **whichever condition has more free antibody has the larger** ρ. Intuitively, free antibody is the "pressure" that pushes antibody onto epitopes. Target epitopes, with their small K_{d1}, are already mostly occupied at low pressure, so raising the pressure helps the weak off-target epitopes more than the target epitopes.

Step 2: Y_f is set implicitly by antibody conservation. Substituting $\theta_1 = Y_f/(Y_f + K_{d1})$ and $\theta_2 = Y_f/(Y_f + K_{d2})$ into antibody conservation $Y = Y_f + S\theta_1 + M\theta_2$ gives

$$Y = F(Y_f; S), \qquad F(y; S) = y + \frac{Sy}{y + K_{d1}} + \frac{My}{y + K_{d2}}. \tag{22}$$

Here $F(y; S)$ is the total antibody that would be needed to produce a free antibody concentration y when the total target is S: y free, plus $Sy/(y + K_{d1})$ bound to target, plus $My/(y + K_{d2})$ bound off-target. We cannot in general solve (22) for Y_f in closed form (after clearing denominators it is a cubic), but we can learn a lot from how F depends on its two arguments.

F is increasing in y. Each fraction has the form $y/(y + K)$, whose derivative by the quotient rule is

$$\frac{d}{dy}\left(\frac{y}{y + K}\right) = \frac{1 \cdot (y + K) - y \cdot 1}{(y + K)^2} = \frac{K}{(y + K)^2}.$$

Applying this with $K = K_{d1}$ and $K = K_{d2}$,

$$\frac{\partial F}{\partial y} = 1 + \frac{SK_{d1}}{(y + K_{d1})^2} + \frac{MK_{d2}}{(y + K_{d2})^2} \geq 1 > 0. \tag{23}$$

(The symbol ∂ means we differentiate with respect to y while holding S fixed.) Also $F(0; S) = 0$ and $F(y; S) \geq y$, so F climbs continuously from 0 to arbitrarily large values. By the intermediate value theorem, for any $Y > 0$ there is a solution Y_f of $F(Y_f; S) = Y$, and because F is strictly increasing in y, it is the only one. Moreover, $Y_f > 0$, since $F(0; S) = 0 < Y$.

F is increasing in S. S appears only in the middle term, so

$$\frac{\partial F}{\partial S} = \frac{y}{y + K_{d1}} = \theta_1 > 0 \quad \text{for } y > 0.$$

More antibody is needed to reach a given free concentration when there is more target to soak it up.

Step 3: smaller S gives larger Y_f. Let Y_f and Y_f' be the solutions for the control (S_c) and treatment (S_t), with the same Y, and suppose $A' < A$, so $S_t < S_c$. Evaluate the treatment's function at the control's free antibody. Since only the middle term depends on S,

$$F(Y_f; S_t) = F(Y_f; S_c) - \frac{(S_c - S_t)\, Y_f}{Y_f + K_{d1}} = Y - (S_c - S_t)\, \theta_1,$$

where we substituted $F(Y_f; S_c) = Y$ from (22). The subtracted term is positive (because $S_c > S_t$ and $\theta_1 > 0$), so

$$F(Y_f; S_t) < Y = F(Y_f'; S_t).$$

Since $F(\,\cdot\,; S_t)$ is increasing (23), a smaller output requires a smaller input, so $Y_f < Y_f'$. In words: removing target epitopes releases the antibody that would have bound them, and the system re-equilibrates at a higher free-antibody concentration.

Step 4: result. Combining Step 3 with (21),

$$A' < A \quad \Longrightarrow \quad Y_f' > Y_f \quad \Longrightarrow \quad \rho' > \rho. \tag{24}$$

This holds in *every* regime, without any approximation. The regime-specific formulas only tell us *how much* larger ρ' is: in the deep limits of regimes 2 and 4 the difference is negligible ($\rho' \approx \rho$), whereas near the crossovers below it can be large.

How large can the difference be? The same calculation can be phrased with implicit differentiation. Differentiating both sides of $Y = F(Y_f(S); S)$ with respect to S (with Y fixed) using the chain rule,

$$0 = \frac{\partial F}{\partial y}\frac{dY_f}{dS} + \frac{\partial F}{\partial S}, \qquad \text{so} \qquad \frac{dY_f}{dS} = -\frac{\partial F/\partial S}{\partial F/\partial y} = -\frac{\theta_1}{\partial F/\partial y}.$$

Since $0 < \theta_1 \le 1$ and $\partial F/\partial y \ge 1$ by (23), we get $-1 \le dY_f/dS < 0$. By the mean value theorem, there is some S^* between S_t and S_c with

$$Y_f' - Y_f = -\left.\frac{dY_f}{dS}\right|_{S^*} (S_c - S_t) \le S_c - S_t = A - A'.$$

That is, removing $A - A'$ target epitopes can raise the free antibody by at most $A - A'$ (the released antibody can only be shared out among solution, the remaining target, and off-target sites). Applying the mean value theorem again, now to ρ as a function of Y_f, and bounding the derivative (21) by its largest value (at $Y_f = 0$),

$$\rho' - \rho \le \frac{K_{d2} - K_{d1}}{K_{d2}^2}\,(Y_f' - Y_f) \le \frac{(1-r)(A - A')}{K_{d2}} < \frac{A - A'}{K_{d2}}. \tag{25}$$

So the *absolute* change in ρ is always modest unless $A - A'$ is comparable to K_{d2}. The *relative* change ρ'/ρ can nevertheless be large when ρ itself is small ($\rho \sim r$) and Y_f is very sensitive to S, which is exactly the situation near equivalence.

2.6.2 Equivalence crossover (between regimes 1 and 3)

Regime 1 required $\sqrt{K_{d1}Y} \ll S - Y$, and regime 3 requires the antibody to exceed the target. Between them lies a window around $Y \approx S$ where antibody and target *titrate* each other. Define the excess target

$$\delta = S - Y,$$

which is positive on the regime-1 side and negative on the regime-3 side.

Assumptions.

1. Off-target-bound antibody is negligible compared with free antibody: $M\theta_2 \ll Y_f$. Since $M\theta_2 = MY_f/(Y_f + K_{d2}) \leq MY_f/K_{d2}$, a sufficient condition is $M \ll K_{d2}$. (We relax this in a remark below.)
2. Target binding is tight compared with the amount of target: $K_{d1} \ll S$.
3. We are near equivalence: $|\delta| \ll S$.
4. Off-target epitopes are far from half-occupied: $Y_f \ll K_{d2}$.

Step 1: an exact quadratic. Dropping $M\theta_2$ (assumption 1) from (1), the free target is

$$S(1 - \theta_1) = (S - Y) + Y_f = \delta + Y_f,$$

and solving antibody conservation for the bound target gives $S\theta_1 = Y - Y_f$. The mass-action relation for target, $K_{d1} = Y_f(A_f + B_f)/(Y_A + Y_B)$, has free target $A_f + B_f = S(1 - \theta_1)$ and bound target $Y_A + Y_B = S\theta_1$. Substituting both and multiplying through by $Y - Y_f$,

$$K_{d1}(Y - Y_f) = Y_f(\delta + Y_f).$$

Expanding and collecting everything on one side,

$$Y_f^2 + (\delta + K_{d1})Y_f - K_{d1}Y = 0. \tag{26}$$

Given assumption 1, this is exact; we will reuse it for the weak-binding crossover.

Step 2: simplify near equivalence. Two terms in (26) are small corrections near equivalence. First, substituting $Y = S - \delta$,

$$K_{d1}Y = K_{d1}S\left(1 - \frac{\delta}{S}\right) \approx K_{d1}S$$

by assumption 3. Second, comparing $K_{d1}Y_f$ with $K_{d1}Y$, their ratio is Y_f/Y, which is $\ll 1$ provided $Y_f \ll Y \approx S$ (we verify this in Step 4). Dropping $K_{d1}Y_f$ and replacing $K_{d1}Y$ by $K_{d1}S$,

$$Y_f\,(Y_f + \delta) \approx K_{d1}S. \tag{27}$$

This has a direct interpretation. $Y_f + \delta$ is the free target, and nearly all of the $\approx S$ target epitopes are bound, so (27) is just mass action: (free antibody)×(free target) = K_{d1}×(bound target).

Step 3: solve the quadratic. Writing (27) as $Y_f^2 + \delta Y_f - K_{d1}S = 0$, the quadratic formula gives

$$Y_f = \frac{-\delta \pm \sqrt{\delta^2 + 4K_{d1}S}}{2}.$$

The product of the two roots equals the constant term, $-K_{d1}S < 0$, so one root is positive and one is negative. Free antibody must be positive, so we take the + sign. Multiplying the numerator and denominator by the conjugate $\delta + \sqrt{\delta^2 + 4K_{d1}S}$ and using $(\sqrt{\delta^2 + 4K_{d1}S})^2 - \delta^2 = 4K_{d1}S$ gives an equivalent form that is convenient when $\delta > 0$:

$$Y_f = \frac{-\delta + \sqrt{\delta^2 + 4K_{d1}S}}{2} = \frac{2K_{d1}S}{\delta + \sqrt{\delta^2 + 4K_{d1}S}}. \tag{28}$$

The natural scale is $\sqrt{K_{d1}S}$. Writing $\delta = x\sqrt{K_{d1}S}$ and dividing (28) by $\sqrt{K_{d1}S}$,

$$\frac{Y_f}{\sqrt{K_{d1}S}} = h(x) = \frac{-x + \sqrt{x^2 + 4}}{2} = \frac{2}{x + \sqrt{x^2 + 4}}. \tag{29}$$

Comparing the two forms of h, $h(-x) = (x + \sqrt{x^2 + 4})/2 = 1/h(x)$: excess target and excess antibody of the same size shift Y_f by the same factor in opposite directions.

Step 4: the three limits.

- *Excess target, $\delta \gg \sqrt{K_{d1}S}$ (regime 1).* Then $4K_{d1}S \ll \delta^2$, so $\sqrt{\delta^2 + 4K_{d1}S} \approx \delta$. Substituting into the second form of (28),

  $$Y_f \approx \frac{2K_{d1}S}{\delta + \delta} = \frac{K_{d1}S}{\delta} = \frac{K_{d1}S}{S - Y}.$$

  Since $S \approx Y$ (assumption 3), this matches the regime-1 estimate $K_{d1}Y/(S - Y)$ of (4). (The exact quadratic (26) gives $K_{d1}Y/(S - Y + K_{d1})$ in this limit, matching even more closely.)

- *Exact equivalence, $\delta = 0$.* Substituting $\delta = 0$ into (28),

  $$Y_f = \frac{\sqrt{4K_{d1}S}}{2} = \sqrt{K_{d1}S},$$

  the geometric mean of K_{d1} and S, as anticipated in the discussion of regime 1's assumption 2.

- *Excess antibody, $-\delta \gg \sqrt{K_{d1}S}$ (regime 3).* Now $\delta = -|\delta|$ and again $\sqrt{\delta^2 + 4K_{d1}S} \approx |\delta|$. Substituting into the first form of (28),

  $$Y_f \approx \frac{|\delta| + |\delta|}{2} = |\delta| = Y - S.$$

  The target is saturated, and the excess antibody is free (it has nowhere else to go, since off-target binding is negligible).

We can now verify the claim $Y_f \ll S$ used in Step 2. Using $\sqrt{a + b} \le \sqrt{a} + \sqrt{b}$ for $a, b \ge 0$ in the first form of (28),

$$Y_f \le \frac{-\delta + |\delta| + 2\sqrt{K_{d1}S}}{2} \le |\delta| + \sqrt{K_{d1}S}.$$

Here $|\delta| \ll S$ by assumption 3, and $\sqrt{K_{d1}S} \ll S$ because, dividing by $\sqrt{S}$, it is equivalent to $\sqrt{K_{d1}} \ll \sqrt{S}$, i.e., assumption 2. So $Y_f \ll S$.

Step 5: ρ. By assumption 4, $Y_f + K_{d2} \approx K_{d2}$, so

$$\rho = \frac{Y_f + K_{d1}}{Y_f + K_{d2}} \approx \frac{Y_f + K_{d1}}{K_{d2}} = r\left(1 + \frac{Y_f}{K_{d1}}\right), \tag{30}$$

where the last step factors out $K_{d1}/K_{d2} = r$. Substituting the three limits of Step 4:

$$\rho \approx \begin{cases} r\left(1 + \dfrac{S}{S-Y}\right) \approx r\dfrac{S}{S-Y}, & S - Y \gg \sqrt{K_{d1}S} \quad \text{(regime 1)}, \\ r\left(1 + \sqrt{\dfrac{S}{K_{d1}}}\right) \approx \dfrac{\sqrt{K_{d1}S}}{K_{d2}}, & Y = S, \\ r + \dfrac{Y-S}{K_{d2}}, & Y - S \gg \sqrt{K_{d1}S} \quad \text{(regime 3, } M \ll K_{d2}\text{)}. \end{cases}$$

(In the first line, $S/(S-Y) \gg 1$ near equivalence, so the 1 is negligible; in the second, $\sqrt{S/K_{d1}} \gg 1$ by assumption 2.) Through the window, ρ climbs from about $r\,S/(S-Y)$ to about $(Y-S)/K_{d2}$, and the climb happens over a range of Y of width a few times $\sqrt{K_{d1}S}$, i.e., a *relative* width of order $\sqrt{K_{d1}S}/S = \sqrt{K_{d1}/S} \ll 1$. For example, by (29), moving from $x = 2$ to $x = -2$ multiplies Y_f by $h(-2)/h(2) = 1/h(2)^2 = (1+\sqrt{2})^2 \approx 5.8$; moving from $x = 4$ to $x = -4$ multiplies it by $\left((4+\sqrt{20})/2\right)^2 \approx 18$.

ρ' versus ρ. The treatment has $\delta_t = S_t - Y = \delta_c - (A - A') < \delta_c$. In (28) (second form), the denominator $\delta + \sqrt{\delta^2 + 4K_{d1}S}$ is increasing in δ, since its derivative is $1 + \delta/\sqrt{\delta^2 + 4K_{d1}S} > 0$ (because $|\delta| < \sqrt{\delta^2 + 4K_{d1}S}$). So decreasing δ increases Y_f, and by (30), $\rho' > \rho$, as guaranteed in general by (24). Because $\rho \propto Y_f + K_{d1}$ here, the ratio is

$$\frac{\rho'}{\rho} \approx \frac{Y_f' + K_{d1}}{Y_f + K_{d1}},$$

which is a ratio of free-antibody concentrations. If the drop $A - A'$ is large enough to carry the treatment across the window while the control stays on the regime-1 side, this ratio is approximately $(Y - S_t)/\big(K_{d1}S_c/(S_c - Y)\big)$, which can be enormous because K_{d1} is small. Equivalently, bound (25) gives $\rho'/\rho \le 1 + (A - A')/(Y_f + K_{d1})$, which is large when $A - A'$ exceeds the free antibody $\sim \sqrt{K_{d1}S}$.

Numerical illustration. Take $K_{d1} = 0.1$ nM, $K_{d2} = 1000$ nM ($r = 10^{-4}$), $A = 90$ nM, $B = 10$ nM, $M \ll K_{d2}$, and a treatment that removes about 11% of the target, $A' = 80$ nM. Then $S_c = 100$ nM and $S_t = 90$ nM, and $\sqrt{K_{d1}S} \approx 3$ nM.

- With $Y = 95$ nM, $\delta_c = 5$ nM and $\delta_t = -5$ nM. Substituting into (28),

$$Y_f \approx \frac{-5 + \sqrt{25 + 4(0.1)(100)}}{2} = \frac{-5 + 8.06}{2} \approx 1.53 \text{ nM},$$
$$Y_f' \approx \frac{5 + \sqrt{25 + 4(0.1)(90)}}{2} = \frac{5 + 7.81}{2} \approx 6.41 \text{ nM}.$$

  By (30), $\rho \approx 1.63/1000 = 1.6 \times 10^{-3}$ and $\rho' \approx 6.51/1000 = 6.5 \times 10^{-3}$, so $\rho'/\rho \approx 4$. (Solving (22) numerically gives $\rho'/\rho \approx 4.2$.)

- With $Y = 50$ nM (deep in regime 1), the same 11% drop gives $\rho'/\rho \approx \dfrac{90/40}{100/50} = 1.125$.

An 11% change in target abundance produces a fourfold change in the relative off-target capture efficiency when the antibody is titrated near equivalence, but only a 12.5% change when the antibody is at half the target concentration.

Remark: an off-target sink. If M is not small compared with K_{d2} but off-target sites are still unsaturated ($Y_f \ll K_{d2}$), then $M\theta_2 \approx MY_f/K_{d2}$, and the antibody not on target is

$$U = Y_f + M\theta_2 \approx gY_f, \qquad g = 1 + \frac{M}{K_{d2}}.$$

Repeating Step 1 with U in place of Y_f in (1) and in antibody conservation, the free target is $\delta + U$ and the bound target is $Y - U$, so mass action reads $K_{d1}(Y - U) = Y_f(\delta + U)$. Multiplying both sides by g and substituting $gY_f = U$,

$$gK_{d1}(Y - U) = U(\delta + U).$$

This is (26) with $Y_f \to U$ and $K_{d1} \to gK_{d1}$. All the results above carry over with these replacements; in particular, $U \approx \sqrt{gK_{d1}S}$ at equivalence and $Y_f = U/g$. The off-target pool acts as a buffer that soaks up a fixed fraction $1 - 1/g$ of the antibody not on target, and it widens the window to $\sqrt{gK_{d1}S}$, which makes the crossover less sharp.

2.6.3 Off-target half-saturation crossover (between regimes 3 and 4)

Once $Y - S \gg \sqrt{K_{d1}S}$, the target is saturated and the remaining question is how the excess antibody distributes between solution and off-target epitopes. Define the excess antibody

$$E = Y - S > 0.$$

Assumption. The target is saturated: the free target $S(1 - \theta_1) = SK_{d1}/(Y_f + K_{d1})$ is negligible compared with E. In this crossover $Y_f \sim K_{d2} \gg K_{d1}$, so the free target is about $SK_{d1}/K_{d2} = rS$, and the assumption amounts to $rS \ll Y - S$.

Step 1: a quadratic for Y_f. Saturation means $S\theta_1 = S - S(1 - \theta_1) \approx S$. Substituting this and $\theta_2 = Y_f/(Y_f + K_{d2})$ into antibody conservation, $Y = Y_f + S + MY_f/(Y_f + K_{d2})$, and subtracting S from both sides,

$$E = Y_f + \frac{MY_f}{Y_f + K_{d2}}.$$

In words, the excess antibody is either free or bound off-target. Multiplying both sides by $Y_f + K_{d2}$,

$$EY_f + EK_{d2} = Y_f^2 + K_{d2}Y_f + MY_f,$$

and collecting terms on one side,

$$Y_f^2 + bY_f - EK_{d2} = 0, \qquad b = K_{d2} + M - E. \tag{31}$$

Step 2: solve. The product of the roots is $-EK_{d2} < 0$, so there is exactly one positive root. By the quadratic formula, and rationalizing as in (28),

$$Y_f = \frac{-b + \sqrt{b^2 + 4EK_{d2}}}{2} = \frac{2EK_{d2}}{b + \sqrt{b^2 + 4EK_{d2}}}, \qquad \rho = \frac{Y_f + K_{d1}}{Y_f + K_{d2}} \approx \frac{Y_f}{Y_f + K_{d2}}, \tag{32}$$

where the last approximation uses $K_{d1} \ll Y_f$.

Step 3: the limits.

- *Small excess, $E \ll K_{d2}+M$ (regime 3).* Then $b \approx K_{d2}+M > 0$ and $4EK_{d2} \le 4E(K_{d2}+M) \ll b^2$, so $\sqrt{b^2 + 4EK_{d2}} \approx b$. Substituting into the second form of (32),

  $$Y_f \approx \frac{2EK_{d2}}{2b} \approx \frac{K_{d2}(Y - S)}{K_{d2} + M}.$$

  This is $\ll K_{d2}$, so $\rho \approx (Y_f + K_{d1})/K_{d2}$; substituting Y_f gives $\rho \approx r + (Y - S)/(K_{d2} + C + D)$, the regime-3 result.

- *Large excess, $E \gg K_{d2} + M$ (regime 4).* Now $b = -(E - M - K_{d2})$ is large and negative. Factoring out $|b|$ and using the linear approximation $\sqrt{1+u} \approx 1 + u/2$ for small u,

  $$\sqrt{b^2 + 4EK_{d2}} = |b|\sqrt{1 + \frac{4EK_{d2}}{b^2}} \approx |b| + \frac{2EK_{d2}}{|b|} \approx |b| + 2K_{d2},$$

  where the last step uses $E/|b| \approx 1$. Substituting into the first form of (32),

  $$Y_f \approx \frac{|b| + |b| + 2K_{d2}}{2} = |b| + K_{d2} = E - M = Y - S - (C + D).$$

  All epitopes are saturated and the leftover antibody is free. Then $\rho \approx (E - M)/(E - M + K_{d2}) \to 1$ when $E - M \gg K_{d2}$, the regime-4 result.

- *Half-saturation.* $\rho \approx 1/2$ when $Y_f = K_{d2}$, i.e., when off-target epitopes are half occupied. Rather than solving (31), substitute $Y_f = K_{d2}$ directly into $E = Y_f + MY_f/(Y_f + K_{d2})$:

  $$E_{1/2} = K_{d2} + \frac{M}{2}, \qquad \text{i.e.,} \qquad Y = S + K_{d2} + \frac{C + D}{2}.$$

  The antibody must cover the target (S), half of the off-target pool ($M/2$), and leave K_{d2} free.

Two shapes of crossover. The size of M relative to K_{d2} controls how the transition looks.

- If $M \ll K_{d2}$, set $M = 0$ in (31): $Y_f^2 + (K_{d2} - E)Y_f - EK_{d2} = 0$. This factors as $(Y_f - E)(Y_f + K_{d2}) = 0$ (check by expanding: $Y_f^2 + K_{d2}Y_f - EY_f - EK_{d2}$), so $Y_f = E$ and

  $$\rho \approx \frac{(Y - S) + K_{d1}}{(Y - S) + K_{d2}}.$$

  This is a gentle hyperbola: ρ rises from 0.09 to 0.91 as $Y - S$ goes from $0.1K_{d2}$ to $10K_{d2}$, spanning two orders of magnitude.

- If $M \gg K_{d2}$, then (31) has exactly the form of (26) with Y_f unchanged, $K_{d1} \to K_{d2}$, $\delta \to M - E$, and $Y \to E$. So the antibody titrates the off-target pool the way it titrated the target in the equivalence crossover: there is a "second equivalence" at $E \approx M$ with window width $\sqrt{K_{d2}M}$. Half-saturation ($E = K_{d2} + M/2$) occurs before this second equivalence, and by $E = M$ we already have $Y_f \approx \sqrt{K_{d2}M} \gg K_{d2}$, so ρ is close to 1.

ρ' versus ρ. Because the target is saturated, S enters only through $E = Y - S$, and the treatment has $E' = E + (A - A') > E$. Differentiating $E = Y_f + MY_f/(Y_f + K_{d2})$ implicitly with respect to E,

$$1 = \frac{dY_f}{dE}\left(1 + \frac{MK_{d2}}{(Y_f + K_{d2})^2}\right), \qquad \text{so} \qquad 0 < \frac{dY_f}{dE} \leq 1.$$

Thus $Y_f' > Y_f$ and $\rho' > \rho$, as (24) requires. But the effect is small. Near half-saturation, (21) with $Y_f = K_{d2}$ gives $d\rho/dY_f \approx K_{d2}/(2K_{d2})^2 = 1/(4K_{d2})$, so by the chain rule

$$\rho' - \rho \approx \frac{d\rho}{dY_f} \cdot \frac{dY_f}{dE} \cdot (A - A') \leq \frac{A - A'}{4K_{d2}}, \qquad \frac{\rho'}{\rho} \lesssim 1 + \frac{A - A'}{2K_{d2}},$$

using $\rho \approx 1/2$ in the second expression. Unlike near equivalence, ρ is not small here, so a small absolute change in ρ cannot become a large relative change. With the parameters of the earlier illustration and $C = D = 1000$ nM, taking $Y = 2500$ nM gives (numerically) $\rho \approx 0.561$ and $\rho' \approx 0.562$.

2.6.4 Weak-binding crossover (between regimes 1 and 2)

When K_{d1} is comparable to S, neither "all target bound" nor "almost no antibody bound" holds. Clearing the denominators of (22) by multiplying both sides by $(Y_f + K_{d1})(Y_f + K_{d2})$,

$$Y(Y_f + K_{d1})(Y_f + K_{d2}) = Y_f(Y_f + K_{d1})(Y_f + K_{d2}) + SY_f(Y_f + K_{d2}) + MY_f(Y_f + K_{d1}),$$

which is a cubic in Y_f. Its formula is not illuminating, so in practice one solves it numerically (Step 2 of the general principle guarantees a unique positive solution). Two observations remove most of the need to do so.

First, if off-target-bound antibody is negligible ($M \ll K_{d2}$), the derivation of (26) never used $K_{d1} \ll S$, so with $\delta = S - Y$,

$$Y_f = \frac{-(S - Y + K_{d1}) + \sqrt{(S - Y + K_{d1})^2 + 4K_{d1}Y}}{2}, \qquad \rho = \frac{Y_f + K_{d1}}{Y_f + K_{d2}}$$

is exact across the whole crossover. (If M is not negligible but off-target sites are unsaturated, the remark on the off-target sink applies.)

Second, the regime-2 limit follows directly from conservation. Since $\theta_1 = Y_f/(Y_f + K_{d1}) \leq Y_f/K_{d1}$ and $Y_f \leq Y$, the antibody bound to target satisfies

$$Y - Y_f \approx S\theta_1 \leq \frac{SY_f}{K_{d1}} \leq \frac{S}{K_{d1}}\,Y,$$

which is $\ll Y$ when $S \ll K_{d1}$, giving $Y_f \approx Y$. As S/K_{d1} grows toward 1, a growing fraction of antibody is captured by the target, Y_f drops below Y, and ρ drops from $(Y + K_{d1})/(Y + K_{d2})$ toward regime 1's $r\,S/(S-Y)$ (if $Y < S$) or toward the equivalence or regime-3 behavior (if $Y \gtrsim S$). Throughout, (24) still guarantees $\rho' > \rho$ when $A' < A$. The difference is small near regime 2: from the implicit derivative in the general principle, $|dY_f/dS| \le \theta_1 \le Y_f/K_{d1}$, which is small when $Y_f \ll K_{d1}$.

2.6.5 ChIP normalization

From Regime 1's ChIP normalization, scaling by total spike-in reads (ChIP-Rx) gives

$$\frac{n_i'/N_s'}{n_i/N_s} = \frac{a_i'}{a_i} \cdot \beta, \qquad \beta = \frac{B + \rho D}{B + \rho' D}.$$

That derivation used only the read model, not any regime-specific approximation for ρ, so it holds everywhere.

The bias never favors the treatment. By (24), $\rho' > \rho$ whenever $A' < A$. Multiplying by $D \ge 0$ and adding B,

$$B + \rho' D \ge B + \rho D > 0,$$

so dividing both sides by $B + \rho' D$ gives $\beta \le 1$, with equality only if $D = 0$ (or approximately, in the deep limits where $\rho' \approx \rho$). The measured fold change is the true fold change multiplied by $\beta \le 1$. For a global decrease, $a_i'/a_i < 1$, so the measured value is even smaller: **ChIP-Rx with total spike-in reads never understates, and generally overstates, a global decrease**, in every regime. (Equivalently, every locus in the treatment is biased downward by the same factor β.)

How large can the bias be? Compare β with ρ/ρ'. Putting both over a common denominator,

$$\beta - \frac{\rho}{\rho'} = \frac{\rho'(B + \rho D) - \rho(B + \rho' D)}{\rho'(B + \rho' D)} = \frac{\rho' B + \rho\rho' D - \rho B - \rho\rho' D}{\rho'(B + \rho' D)} = \frac{B(\rho' - \rho)}{\rho'(B + \rho' D)} \ge 0.$$

Therefore

$$\frac{\rho}{\rho'} \le \beta \le 1.$$

The bias approaches its worst case ρ/ρ' when $\rho' D \gg B$, i.e., when most spike-in reads come from off-target spike-in epitopes, and vanishes when $\rho' D \ll B$. Combined with the results above:

- Deep in regimes 1 (with $Y \ll S$), 2, and 4, $\rho' \approx \rho$, so $\beta \approx 1$.
- Near equivalence, ρ'/ρ can be large (fourfold in the illustration), so β can be far below 1 if $\rho' D$ is comparable to or larger than B. This is where the bias is largest.
- Near off-target half-saturation, $\rho'/\rho \lesssim 1 + (A - A')/(2K_{d2})$, so β is close to 1 unless $A - A'$ is comparable to K_{d2}.

Numerical illustration. Use $K_{d1} = 0.1$ nM, $K_{d2} = 1000$ nM, $A = 90$ nM, $A' = 80$ nM, $B = 10$ nM, and now a substantial off-target pool $C = D = 1000$ nM (so $g = 3$ in the off-target-sink remark). The true global fold change is $80/90 \approx 0.89$. Solving (22) numerically:

- $Y = 50$ nM (regime 1): $\beta \approx 0.998$.
- $Y = 95$ nM (near equivalence): $\rho \approx 1.20 \times 10^{-3}$, $\rho' \approx 2.82 \times 10^{-3}$, so $\rho D \approx 1.2$ nM and $\rho' D \approx 2.8$ nM, giving $\beta \approx 11.2/12.8 \approx 0.87$. The measured fold change is $0.89 \times 0.87 \approx 0.78$. Scanning Y, the smallest β (≈ 0.83) occurs at $Y \approx 107$ nM, just past equivalence (the off-target sink shifts it slightly).
- $Y = 2500$ nM (off-target half-saturation): $\beta \approx 0.998$.

As in Regime 1, counting only reads at the spike-in's target loci ($N_{s,B}$) removes the dependence on ρ and ρ' entirely, and therefore removes this bias in every regime.